

Master Thesis

Leveraging Generative LLMs and Encoder Models for Multi-Label Topic Classification of Dutch Citizens' Reports in the Governance Domain

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Abstract

This thesis addresses the multi-label topic classification of Dutch citizens' reports in the governance domain, in a context of scarcity of annotated data: the reports do not have labels validated by human annotators, but only those generated by a rule-based system used within the partner company. Generative language models can produce such labels automatically, but how well an encoder trained on them generalizes, compared to using the generative model directly, has been little explored for multi-label topic classification in the governance domain.

To address this problem, a generative model (GPT-4.1) is used to automatically annotate the training data, on which an encoder (BERTje) is then trained. Three approaches are compared: a BERTje trained on rule-based system labels ($\text{BERTje}_{\text{RB}}$), the zero-shot prediction of two generative models (GPT-4.1 and Llama 3.3 70B Instruct), and a BERTje trained on the labels generated by one of the generative models ($\text{BERTje}_{\text{LLM}}$). The evaluation is conducted on a human-validated test set and on a re-annotated subset of it, less dependent on the rule-based system. BERTje trained on labels generated by the generative language model achieves performance comparable to GPT-4.1 zero-shot predictions (micro-F1 of 0.78 versus 0.77 on the main topics and micro-f1 OF 0.46 against 0.48 for the subtopics on the human-validated test set, and results that are nearly identical on the re-annotated subset), despite being a smaller model and without any manually annotated training data. Given equivalent performance, the encoder is the preferred option for operational use, as it can be run locally, at a lower cost, and without sending sensitive data to external services.

Declaration of Authorship

I, Gianmarco Sapienza, declare that this thesis, titled *Leveraging Generative LLMs and Encoder Models for Multi-Label Topic Classification of Dutch Citizens' Reports in the Governance Domain* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: June 26, 2026

Signed: 

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Chapter 1

Introduction

Public institutions provide services to citizens and regularly gather their observations. For these institutions, and in particular for municipalities, knowing what citizens communicate (the problems they report, the requests they make, and the questions they raise about the neighborhood and local services) is an essential function, because it allows resources to be directed towards the actual needs of the population and to respond in a timely manner. This communication can take different forms. Feedback can be actively solicited through surveys, but a significant part of it is also spontaneous: it is the citizen who initiates the contact, about a specific problem, through heterogeneous channels such as email or official website forms. This results in a diversified body of citizens' reports, written in varied and unstructured language, in which the same issues can be expressed in very different ways.

Citizens' reports accumulate in large volumes. To make use of them and identify recurring themes, they must be classified according to their content. The manual analysis of such a high volume of text is time-consuming and necessitate substantial resources. Instead, automated approaches offer a scalable way to extract this information. Extracting information from unstructured textual data is a complex task, for which Natural Language Processing (NLP) can offer a solution.

In particular, NLP is employed to address the multi-label topic classification (MLTC) of Dutch citizens' reports in the governance domain. The task is addressed in a context of scarcity of annotated data: the training data provided by the partner company are not annotated, while a separate, human-validated test set is available for evaluation. The reports concern themes such as waste collection, parking, street maintenance, and public safety. For instance, a report reading *"Fireworks have caused damage to the bus stop. The glass has been shattered on both sides."* concerns both safety and damages claims, and would therefore receive more than one label. This problem can be addressed thanks to the recent development of generative language models, which can perform tasks without task-specific training.

More precisely, this work compares two approaches for handling MLTC: direct zero-shot prediction with generative language models, and the fine-tuning of an encoder model (BERTje, (de Vries et al., 2019)) on labels generated by a generative language model (LLM-predicted labels). The experiments are conducted on data provided by the partner company, where the training set is unannotated and only the test set has been human-validated. The central difficulty is therefore to identify the best approach when no annotated training data are available in the specific domain.

1.1 Problem Description

The reports analyzed in this work are provided by MarketResponse[1]. The company processes citizens' reports from municipalities across the country. Upon arrival, the reports are anonymized and automatically labeled by a rule-based system following a hierarchical taxonomy of main topics and subtopics (see Section 3.1). A single report can concern multiple categories simultaneously: it may, for example, concern both *cleaning* and *bulky waste*, or *maintenance* and *safety*. Classifying each report according to its topic allows the municipality to identify recurring themes and to route each report to the authority responsible for handling it. In case of misclassification, the processing of a citizen's request may slow down.

Currently, the company relies on a rule-based system that assigns labels through hand-crafted rules. This system is fragile: it fails to classify reports whose phrasing is not encoded in its rules, and it does not generalize to cases the rules do not cover. This limitation is increased by the difficulties typical of user-generated text: the reports are spontaneous and informal, contain spelling and grammatical errors, and are at times ambiguous. The distribution of the topics is also highly imbalanced, with one category, *cleaning, waste and maintenance*, more frequent than the others. This imbalance tends to bias classifiers towards the majority class, at the expense of the minority classes (see Chapter 3).

The data are in Dutch and are processed in their original language, so that no information is lost in the process.

The constraint of this work is the scarcity of annotated data. No annotations are available for the training data, and producing them manually would be expensive and time-consuming, especially given the large number of classes and the specialized domain. Generative language models can perform tasks without task-specific training, through prompting techniques such as zero-shot prompting, and have been shown to be a viable alternative to manual annotation (Ding et al., 2023), offering a way to address the scarcity. They can either classify reports directly in a multi-label format or generate LLM-predicted labels with which to train another model. A central question of this work focuses to whether to use generative language models for zero-shot prediction, or to generate LLM-predicted labels with which to fine-tune a smaller encoder model.

1.2 Task description: Multi-label Topic Classification

The goal of multi-label topic classification is to assign multiple topics to a single text. Each instance is associated with a subset of labels, rather than a single label (Herrera et al., 2016). For this reason, for any instance x, i.e., a text document, the output is defined as $Y \subseteq \mathcal{Y}$, where Y is the predicted subset of labels for that instance and $\mathcal{Y}$ represents the entire label space $\{y_1, y_2, \ldots, y_n\}$ (Zhang and Zhou, 2014). When analyzing citizens' reports within the governance domain, a single report may be associated with combinations of topics and/or subtopics, for instance "Municipal Taxes" and "Energy Costs" for the same report.

Handling multi-label instances requires either a specific model architecture or an adaptation of the data, because standard algorithms are not designed to predict multiple labels. To address multi-label classification, Tsoumakas and Katakis (2007) distinguish between two main categories: *problem transformation* and *algorithm adaptation*.

[1] marketresponse.nl

The former transforms the task into one or more single-label classification tasks. Some of these problem transformation methods may ignore label correlations or augment the label space (Tsoumakas and Katakis, 2007). The latter directly modifies the model architecture to output multiple labels (Tsoumakas and Katakis, 2007). Single-label classification typically refers to binary classification and multi-class classification. Additionally, ensembles of classifiers (Herrera et al., 2016) combine the predictions of multiple base classifiers to capture label correlations, thereby improving overall predictive performance.

1.3 Research Question(s)

The research questions of this thesis are organized into one main question and two subquestions, formulated as follows.

Main Question: Which approach yields the best performance for multi-label topic classification in the governance domain: direct zero-shot prediction using generative language models or an encoder model fine-tuned on LLM-predicted labels?

Subquestion 1: What is the predictive performance of zero-shot generative language models for multi-label topic classification on Dutch governance data when evaluated on a human-validated test set?

Subquestion 2: What is the predictive performance of a transformer-based encoder fine-tuned on LLM-predicted labels for multi-label topic classification on Dutch governance data when evaluated on the same human-validated test set?

1.4 Thesis Outline

This work compares two approaches to MLTC of Dutch citizens' reports: zero-shot prediction with generative language models, and the fine-tuning of an encoder model on LLM-predicted labels. The generative language models are GPT-4.1 (OpenAI et al., 2024) and Llama 3.3 70B Instruct (Grattafiori et al., 2024), and the encoder is BERTje (de Vries et al., 2019), a BERT model pre-trained for Dutch. Both approaches are evaluated against the human-validated test set provided by the partner company. The full experimental design is described in Chapter 4.

This thesis is organized as follows. Chapter 2 reviews related work on MLTC: early machine learning approaches, transfer learning, and the use of generative language models for classification. Chapter 3 describes the data, together with the cleaning and balancing procedures applied to it. Chapter 4 describes the classification methods and the experimental setup. Chapter 5 presents the results of the experiments, together with an error analysis of the best-performing model. Chapter 6 discusses the results and presents the conclusions of this work, including its limitations and possible directions for future work.

Chapter 2

Related Work

This chapter will provide a comprehensive review of the related work concerning Multi-Label Topic Classification (MLTC), with a focus on approaches and the emerging use of Large Language Models (LLMs) to predict labels for datasets lacking annotations. This thesis will focus on prediction methods that incorporate transfer learning techniques. A brief overview of traditional machine learning will be provided; however, this approach will not be considered in this research. The chapter will mainly focus on methodologies in which transformer-based models such as BERT have been used for MLTC, before focusing on the use of generative language models for multi-label topic classification. To train the BERT model, LLM-predicted labels will be used. For this reason, this chapter will also review techniques that employ such labels as a training data.

2.1 Traditional Machine Learning

Traditional Machine Learning (ML) includes algorithms such as Naive Bayes, Support Vector Machines, and Logistic Regression. These methods require the manual extraction of text characteristics to model language; for this reason, they are also referred to as "feature-based" models (Jurafsky and Martin, 2026).However, these algorithms present limitations when applied to multi-label topic classification, as they only deal with binary or multi-class classification (Herrera et al., 2016) and are not designed to assign multiple labels to a single instance. They are not designed to handle several categories simultaneously, meaning their output does not consider the overlapping of multiple topics. Consequently, to use traditional ML methods for multi-label tasks, researchers had to apply problem transformation or algorithm adaptation techniques (Zhang and Zhou, 2014) (previously mentioned in the Section 1.2). Traditional ML approaches present an additional limitation: their high dependency on manual feature engineering. The quality of the classification is heavily influenced by the quality of the characteristics extracted from the text during preprocessing. Aggarwal and Zhai (2012) underline this feature selection problem, pointing out that different methods propose different features according to the selected task.

In contrast, ML models based on deep learning learn representations through dense, low-dimensional vectors, significantly reducing the necessity for hand-crafted features. As pointed out in (Chen et al., 2022), modern deep learning architectures extract features from text autonomously. Despite this shift, traditional ML models for MLTC are extensively documented (Shah et al., 2020; Zhang and Zhou, 2014) and cover various adaptation methods. For example, algorithm adaptation is widely researched in tradi-

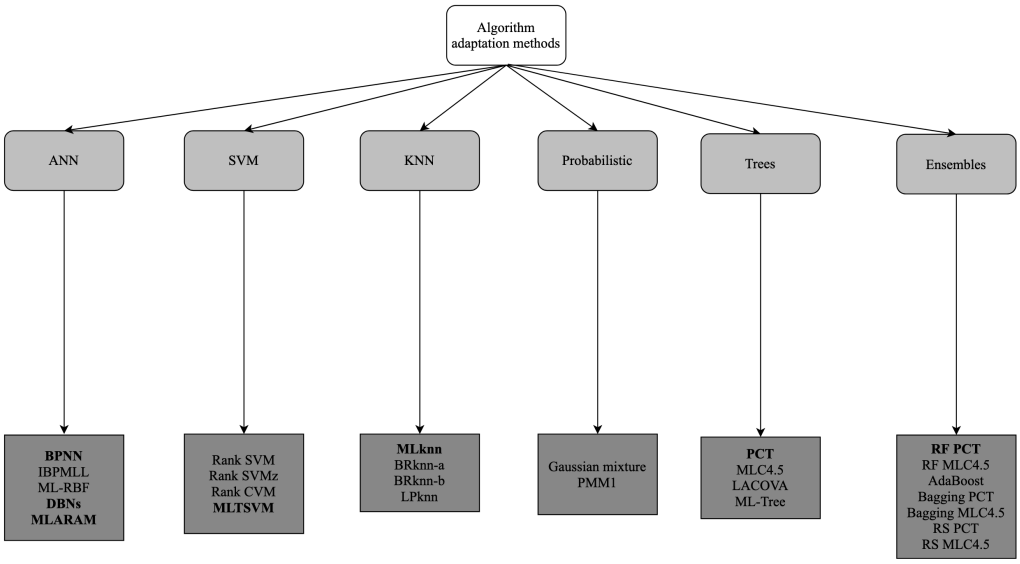


Figure 2.1: Overview of algorithm adaptation as shown in (Bogatinovski et al., 2021).

tional ML, leading to well-established algorithms adapted for MLTC, such as MLTSVM and MLKNN (see Figure 2.1) (Bogatinovski et al., 2021) which are adaptation of handling multi-label data. As noted in (Bogatinovski et al., 2021), adaptation occurs at different algorithmic levels, particularly at the output level by modifying the probability estimation or the cost function optimization (for ML-KNN or the different SVM adaptations).

Traditional ML methods will not be utilized in this work, which will focus instead on transfer learning approaches, particularly employing pre-trained Transfromers.

2.2 Transfer Learning

Transfer learning is a widely used technique in Machine Learning. It becomes particularly useful when performing a task for which extensive data is not available, since it makes it possible to adapt existing models, possibly developed for other languages or tasks, to a different target (Wang and Zheng, 2015). Although it is a well-established technique, the advent of Transformer models such as GPT and BERT marks new developments. These models are pre-trained on more general tasks, and fine-tuning allows them "to converge faster and with relatively lower requirements of fine-tuning data" (Malte and Ratadiya, 2019). They have been increasingly applied to text classification, including MLTC, redefining the state-of-the-art (Vajjala and Shimangaud, 2025).

Pre-trained Transformers are particularly valuable in settings where labeled data are scarce, since even a small amount of annotation can produce effective results, as shown in the experiments by Gao et al. (2020), where fine-tuning on a limited number of examples is evaluated on classification problems such as sentiment analysis and Natural Language Inference. This property is central to the present work, in which annotated training data are not available. The reshaping of NLP tasks brought by the Transformer architecture is largely due to this ability to adapt general pre-trained knowledge to specific downstream tasks (Kalyan et al., 2021).

Among pre-trained Transformers, two architectural families are particularly relevant to this work: encoder-based models, such as BERT, and decoder-based generative models, such as GPT. The following sections review how each family has been applied to multi-label topic classification.

2.2.1 BERT for Multi-Label Topic Classification

BERT (Bidirectional Encoder Representations from Transformers) has been successfully applied to multi-label classification tasks on user-generated content. Schonlau et al. (2023) employ BERT for the multi-label classification of open-ended survey questions, evaluating which classification method minimises the 0/1 loss. Empirical results show that BERT achieves the lowest error rate when compared to classical problem transformation approaches (Binary Relevance, Ensemble Classifier Chains and Label Powerset). The same study examines texts characterised as "mildly multi-label", in which the average number of labels per instance is relatively low. On such instances, the Transformer-based architecture outperforms competitive baselines such as logistic regression. A similar comparison between different models and baselines has also been successfully carried out for multi-label text classification in (Adhikari et al., 2019), where $BERT_{large}$ and $BERT_{base}$ achieve the best results for multi-label document classification. On the two multi-label datasets used in that study, Reuters-21578 and AAPD, both BERT variants obtain the highest F1 ($BERT_{large}$: test F1 of 90.7 on Reuters and 75.2 on AAPD; $BERT_{base}$: test F1 of 89.0 on Reuters and 73.4 on AAPD), improving over a strong SVM baseline (test F1 of 86.1 on Reuters and 69.1 on AAPD) and over the previous state-of-the-art neural model, a regularised BiLSTM (87.0 on Reuters and 70.5 on AAPD), with $BERT_{large}$ performing best overall.

The applicability of BERT extends beyond moderate settings. Chang et al. (2019) address *extreme multi-label text classification*, where the label set can reach hundreds of thousands of classes (for example, the Wikipedia taxonomy), showing that BERT can be adapted and scaled to handle very large label sets. While the present work operates on a substantially smaller label set, and standard fine-tuning is therefore sufficient, these results illustrate the scalability of the architecture across different dataset sizes.

Taken together, these studies demonstrate the effectiveness of Transformer-based models, and of BERT in particular, for MLTC across different scales and settings. Building on these works, the present thesis fine-tunes BERTje, the Dutch BERT model, for MLTC in the governance domain. Since annotated training data are not available in this setting, the training labels are generated by a generative language model. The procedure for generating these LLM-predicted labels is detailed in Section 4.2. The following section first reviews how generative models themselves have been applied to multi-label topic classification.

2.2.2 Generative Language Models for Multi-Label Topic Classification

Generative language models, such as those of the GPT family (Generative Pre-trained Transformers), are decoder-only, instruction-based models. Their primary goal is text generation, but they can be applied to tasks for which they were not explicitly trained through prompting (Brown et al., 2020). Unlike encoder models such as BERT, generative language models do not require task-specific fine-tuning to perform a new task, although fine-tuning remains possible for decoder-only models as well. Through zero-

shot and few-shot prompting, the target task is described in natural language and the model produces predictions without any update to its weights. This makes generative language models practical in settings where labeled data are not available.

Some studies have applied generative language models to multi-label topic classification using zero-shot prompting. Peskine et al. (2023) employ a GPT model to classify multi-label tweets concerning conspiracy theories, showing that a generative language model can assign multiple labels to a text without task-specific training, and that the formulation of the prompt affects the quality of the predictions. This corresponds to direct zero-shot prediction, one of the methods evaluated in this work.

Studies that directly compare generative language models and fine-tuned encoder models specifically on multi-label topic classification are still limited. The available evidence comes mostly from other text classification tasks, where the comparison tends to favour fine-tuned encoder models, despite their smaller size (Wang et al., 2024). This is confirmed by Bucher and Martini (2024), who compare generative language models instructed through zero-shot prompting with smaller encoder models fine-tuned on a range of classification tasks (for example, sentiment analysis, approval/disapproval recognition, emotion detection, and identification of political party positions). Their study finds that fine-tuned encoders outperform zero-shot generative models in all four tasks, with the advantage being most pronounced on the more specialized, non-standard tasks such as emotion detection and party position classification.

This difference is attributable to the prompting mechanism: since the weights are not updated, the internal representations remain unchanged, and performance depends on the capabilities acquired during pre-training (Kong et al., 2024). Generative language models do not necessarily possess specialized knowledge of the target domain or of the task label set, and instruction tuning exhibits documented limitations in generalizing to specific tasks (Naim et al., 2025). These observations do not imply the superiority of encoder models over decoder-only models, or vice versa, but motivate a comparison between the direct zero-shot prediction of generative language models and encoder fine-tuning, which this thesis explores. When the objective is to fine-tune an encoder such as BERTje and no annotations are available, however, a scarcity of annotated labels arises. A recent line of research addresses this problem by using generative language models to generate the training labels automatically, an approach reviewed in the next section.

2.3 Label Scarcity Problem

Encoder models can achieve high performance on classification tasks, but their fine-tuning requires labeled data specific to the task. The development of pre-trained Transfromers has partly mitigated the scarcity of labeled data, without eliminating it: supervised training still depends heavily on the availability of labels (Ding et al., 2023). The manual annotation of these labels, however, is an expensive and time-consuming process, particularly when the classes are numerous and the data come from a specialized domain, and is therefore difficult to gather high quality annotation.

A recent line of research addresses this limitation by employing generative language models to produce training labels automatically. Gilardi et al. (2023) show that ChatGPT surpasses crowd workers on several text-annotation tasks, including topic classification, with higher zero-shot accuracy and at a lower cost. Similarly, Ding et al. (2023) evaluate the use of a generative language model as an annotator of training

data for supervised models, highlighting how the resulting labels can reduce annotation costs compared to manual annotation. These studies indicate that a generative language model can serve as a source of training labels.

On this basis, the present work employs a generative language model to produce LLM-predicted labels with which a smaller encoder model (BERTje) is fine-tuned. It should be noted that the label scarcity problem concerns the fine-tuning of the encoder model exclusively: zero-shot prompting does not require labeled data, and the decoder-only models are not trained. Label scarcity therefore applies only to the fine-tuning of BERTje, and not to direct zero-shot prediction.

In the same domain, Szabo (2024) employs a generative language model to produce synthetic data supporting the balancing of the training set, although that work uses feedback retrieved from survey responses rather than the spontaneous citizens' reports considered here. The present work extends this use of generative language models: rather than focusing on data augmentation, it generates the training labels for the entire training set and uses them to fine-tune an encoder model. It remains to be assessed to what extent LLM-predicted labels allow an encoder model to be trained effectively for Dutch citizens' reports. It should also be considered that labels generated by a generative language model may reflect the biases present in the model's pre-training data (Ding et al., 2023), an aspect relevant to the quality of the resulting training set.

2.4 Governance Domain

Research on MLTC specifically in the governance domain is limited, in particular on citizens' reports. Nugent et al. (2020) further pre-train a BERT model on financial and business text and then fine-tune it for the classification of ESG (Environmental, Social and Governance) topics. ESG concerns governance in a corporate sense, which is distinct from public governance and, in particular, from the relationship between citizens and public administration considered in this work. The reference is relevant here as an example of domain adaptation under conditions of data scarcity. Closer to the context considered in this thesis, Xiong et al. (2024) classify public service requests, using a BERT model to obtain vector representations of the text in combination with other neural components (BiLSTM and CNN). This further confirms the use of encoder models for text produced in the context of public services.

More recently, generative language models have been employed for the multi-class classification of public-service feedback. Rakhimzhanov et al. (2025) apply generative language models to user-submitted content, specifically to the classification of public-transport complaints, a domain close to the one explored in this thesis. They compare prompting-based LLMs with multilingual embedding models that map text in different languages into a shared dense vector space and classify it by similarity. The authors find that the fine-tuned multilingual embedding models match or surpass the accuracy of the LLMs, reaching up to around 90%, while being considerably faster and more computationally efficient. This indicates that smaller, fine-tuned models can be a competitive alternative to generative language models for this type of task.

The closest precedent in this domain is the work of Szabo (2024), conducted on the same governance domain and on similar multi-label data. It compares a fine-tuned BERT model with Support Vector Machines using a two-step approach, and employs a generative language model (GPT-4) to produce synthetic data for oversampling the underrepresented classes. The fine-tuned BERT model trained on the oversampled set is

the best performing configuration, with a macro-averaged F1-score of 0.66, outperforming the SVMs (0.57 and the baseline of 0.56). In that study, however, the generative language model is used only for data augmentation, and not for direct zero-shot prediction or for generating the labels used to train the encoder. To date, no study applies generative language models to MLTC of Dutch citizens' reports in the governance domain, nor uses LLM-predicted labels to fine-tune an encoder model in the same domain. This thesis aims to fill this gap, situating itself at the intersection of MLTC based on transfer learning, automatic label generation through generative language models, and the Dutch governance domain.

2.5 Concluding remarks

This chapter has reviewed the principal strategies for MLTC, from traditional machine learning and its dependence on manually engineered features, to transfer learning and Transformer-based models. It has then presented the use of encoder-only and decoder-only models for classification, the problem of label scarcity, and the limited research currently available in the governance domain.

Taken together, the studies reviewed suggest two directions that have been separate. On one side, generative language models have been shown to be effective annotators, able to produce training labels at a lower cost than manual annotation (Gilardi et al., 2023; Ding et al., 2023). On the other side, when an encoder model is fine-tuned on a classification task, it tends to match or surpass the direct zero-shot prediction of a generative language model, while being smaller and computationally cheaper (Bucher and Martini, 2024; Wang et al., 2024). In a domain close to the present one, Rakhimzhanov et al. (2025) reach the same conclusion for public-transport complaints. In the governance domain specifically, Szabo (2024) showed that a fine-tuned BERT model outperforms an SVM and benefits from GPT-4 generated synthetic data, but the generative model was used only for data augmentation, and not to label the training set or to predict the report labels directly.

What remains open is whether these two directions can be brought together in this specific domain. No prior work has tested, for Dutch citizens' reports in the governance domain, whether labels generated entirely by a generative language model can fine-tune an encoder (BERTje) that matches or surpasses the direct zero-shot prediction of the same models. This is the question this thesis addresses: it compares the direct zero-shot prediction of generative language models (GPT-4.1 and Llama 3.3 70B Instruct) with an encoder model (BERTje) fine-tuned on LLM-predicted labels, against the background of a BERTje fine-tuned rule-based labels. Since no annotations are available for the training set, the training labels are generated by the best-performing generative language model, selected by predicting on the human-validated test set. The selected model, the classification approaches and the experimental design are detailed in Chapter 4.

Chapter 3

Data

The data provided by MarketResponse consist of Dutch citizens' reports from municipalities in the Netherlands. The language of the data is Dutch (ISO 639-1: nl). The reports are sent to the partner company MarketResponse, which anonymizes them and runs a rule-based system that assigns topic labels to each report.

This chapter describes the data. Section 3.1 presents the data collection and preparation. Section 3.2 describes the main topics and subtopics. Section 3.3 describes the cleaning and undersampling of the training data, and Section 3.4 its distribution after preprocessing. Section 3.5 describes the two test sets.

3.1 Data Collection and Preparation

Citizens send their reports to the municipality through different channels: 1) the official website form; 2) email; 3) transcriptions of phone calls. A report is a complaint or a concern raised spontaneously by a citizen. The reports concern issues of urban life, such as litter, vandalism, parking, or noise. The reports are usually short: a report can consist of a single word up to a few sentences. The length statistics are reported in Table 3.1. Each report can have up to three main topics, and each main topic is associated with one or more subtopics. The taxonomy used in this work consists of 10 main topics and 37 subtopics, shown in Figure 3.1. An illustrative example of a report is as follows:

> 'In [PII_LOCATION] there's a huge whole in the ground, I fell into it, very dangerous!"

Before the reports are handed over, they pass through an anonymization system that replaces Personally Identifiable Information (PII) with placeholder tokens, such as `[PII_LOCATION]` in the example above (but also `[PII_NAME]`, `[PII_DATE]`, `[PII_NUMBER]`.

The training data consist of these reports together with the labels assigned to them. Two label sets are assigned to the same reports: the rule-based labels produced by the company's system, and a set of labels generated by GPT-4.1 (the LLM-predicted labels). The preparation of the training set and the two label sets are described in Section 3.3.

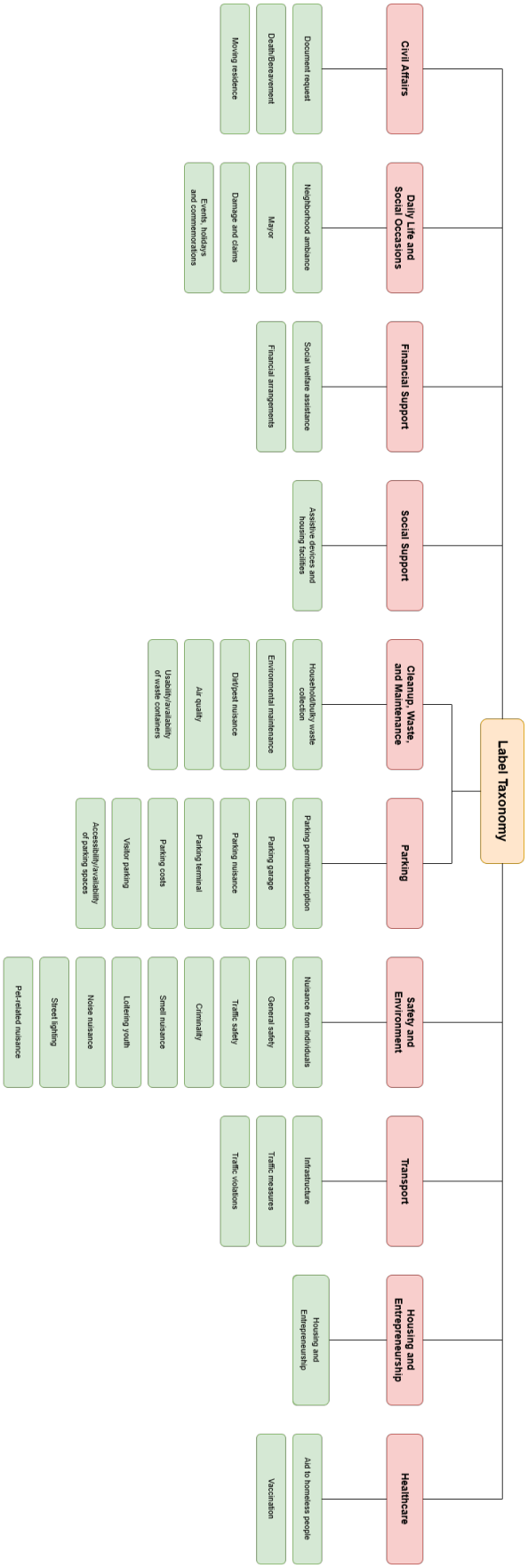


Figure 3.1: The label taxonomy used in the experiment, translated into English.

Table 3.1: Report length statistics for the training set and the test set.

	Training set	**Test set**
Reports	27,732	1,303
Mean words per report	32.50	19.9
min words per report	1	1
Max words in a report	205	172

3.2 Description of Main Topics and Subtopics

The following is a description of the main topics present in this work. In the description, an overall view of what the subtopics cover is given. The description is taken from an official guideline provided by the partner company.

Burgerzaken → Civil Affairs

This topic covers all civil matters, such as marriage, births, funerals, certificates, and any inquiries regarding IDs and passports. Its subtopics are *documentaanvraag* (*document request*), *overlijden* (*death*) and *verhuizen* (*moving house*).

Dagelijks leven en sociale gelegenheden → Daily Life and Social Occasions

This topic mainly covers social events and activities, but it also includes general comments about the overall atmosphere of a particular neighborhood. It is not uncommon to find in this section general comments about social life, reports of damage, or reports of lost and found property (e.g., complaints about bikes that have been parked in wrong places). Its subtopics are *buurt ambiance* (*neighbourhood atmosphere*), *burgemeester* (*mayor*), *schade en claims* (*damage and claims*) and *evenementen, feestdagen en herdenkingen* (*events, holidays and commemorations*).

Financiële ondersteuning → Financial Support

This main topic covers a wide range of financial issues. It includes information on benefit applications, youth allowances, and claims related to fines or other payments, including questions about payment procedures. Its subtopics are *bijstand* (*social assistance*) and *financiële regelingen* (*financial arrangements*).

Maatschappelijke ondersteuning → Social Support

This main topic includes reports from residents who need special assistance, have limited mobility, or face other challenges in their daily lives. These inquiries range from mobility issues (such as whether a particular passageway is inaccessible to wheelchair users) to questions regarding supported living, care coordinators, day care, and therapy. Its subtopic is *hulpmiddelen en woonvoorzieningen* (*aids and housing facilities*).

Opruimen, afval en onderhoud → Cleanup, Waste, and Maintenance

This topic contains reports from residents regarding cleanliness and access to trash containers, and covers various issues related to general neighborhood maintenance. For example, it might include reports such as "bulky waste in the street," as well as "residential waste bin is locked" to indicate that residents are unable to access it. Its subtopics

are *ophalen huiscontainers/grofvuil* (*collection of household containers/bulky waste*), *onderhoud omgeving* (*environment maintenance*), *vuil/ongedierte overlast* (*dirt/pest nuisance*), *luchtkwaliteit* (*air quality*) and *bruikbaarheid/beschikbaarheid afvalcontainers* (*usability/availability of waste containers*).

Parkeren → Parking

This topic primarily covers issues related to parking. Many of the reports concern illegally parked vehicles, but they may also include complaints about parking tickets or access to parking lots. Its subtopics are *parkeervergunning/abonnement* (*parking permit/subscription*), *parkeergarage* (*parking garage*), *parkeeroverlast* (*parking nuisance*), *parkeerterminal* (*parking terminal*), *parkeerkosten* (*parking costs*), *parkeren bezoekers* (*visitor parking*) and *bereikbaarheid/beschikbaarheid parkeerplek* (*accessibility/availability of parking space*).

Veiligheid en omgeving → Safety and Environment

This topic extensively covers reports regarding various concerns about the stability and safety of the neighborhood. It includes subtopics such as *criminality*, involving drug use or vandalism, as well as issues that can lead to a decline in overall safety (such as the absence of streetlights or particularly dark areas). Road safety is also addressed, which may overlap with issues related to mobility. Its subtopics are *overlast personen* (*people nuisance*), *algemene veiligheid* (*general safety*), *verkeersveiligheid* (*traffic safety*), *criminaliteit* (*criminality*), *geuroverlast* (*odour nuisance*), *hangjongeren* (*loitering youths*), *geluidsoverlast* (*noise nuisance*), *verlichting* (*lighting*) and *huisdierenoverlast* (*pet nuisance*).

Vervoer → Transport

This topic covers inquiries regarding road conditions, public services, and traffic violations involving private vehicles. Some of the content in this category may overlap with the "Safety and Environment" topic. Primarily, it addresses infrastructure issues, such as damaged crosswalks or road signs that are illegible, misplaced, or damaged. Its subtopics are *infrastructuur* (*infrastructure*), *verkeersmaatregelen* (*traffic measures*) and *verkeersovertreding* (*traffic violation*).

Wonen en ondernemen → Housing and Entrepreneurship

This topic covers various aspects and questions related to renting and buying real estate, as well as municipal taxes related to households (such as WOZ, OZB, waste disposal fees, and other municipal charges). Its subtopic is *huur en verhuur* (*renting and letting*).

Zorg → Healthcare

This category covers matters related to medical advice, such as referrals, doctors' recommendations, and questions about general health issues. It also includes assistance for the homeless, vaccinations, and inquiries regarding COVID-19. Its subtopics are *hulp aan dak- en thuislozen* (*help for the homeless*) and *vaccinatie* (*vaccination*).

3.3 Data Cleaning and Undersampling

The original corpus provided by the company comprises 39,055 reports, labelled by the rule-based system. Two preparation steps were applied before fine-tuning: the removal of duplicates and the undersampling of the dominant main topic.

First, duplicate texts were removed, reducing the corpus from 39,055 to 36,566 reports (2,489 duplicates removed).

Second, the corpus is heavily imbalanced towards the dominant main topic *Opruimen, afval en onderhoud* (Cleanup, Waste, and Maintenance), which is present in 23,460 of the 36,566 reports (64.16%). To reduce this, the dominant main topic was undersampled. The undersampling was applied only to pure single-label instances of the dominant class, that is, reports whose only label was *Opruimen, afval en onderhoud* (*Cleanup, Waste, and Maintenance*). Removing reports where the dominant topic co-occurs with other topics might also remove instances of less frequent topics (Charte et al., 2015), so these were kept. A total of 8,834 pure single-label instances were removed (with a fixed random seed of 42 for reproducibility), reducing the corpus from 36,566 to 27,732 reports. The motivation for this reduction step is cost: the LLM-predicted labels are generated by sending each report to the GPT-4.1 API, and a smaller corpus lowers the cost of this generation.

After undersampling, the dominant main topic is present in 14,626 of the 27,732 reports (52.74%). The dataset remains imbalanced, but the dominant class is reduced (Table 3.2). Because only pure single-label instances were removed, the counts of the other main topics are unchanged.

The undersampling was carried out on the rule-based labels. The same 27,732 reports were then used to generate the LLM-predicted labels with GPT-4.1. For 67 reports (0.2% of the set), GPT-4.1 returned a malformed or invalid output. These reports were kept rather than removed. Both label sets therefore cover exactly the same 27,732 texts, which allows the two to be compared on identical reports. Under the LLM-predicted labels, the dominant main topic (*Opruimen, afval en onderhoud*) is present in 15,675 of the 27,732 reports (56.52%).

Table 3.2: Frequency of the dominant main topic *Opruimen, afval en onderhoud* before and after undersampling, and under the two label sets. Percentages are calculated on a report level. *n* represents the number of reports.

Label set	**n**	**% of reports**
Rule-based labels, before undersampling (36,566 reports)	23,460	64.16
Rule-based labels, after undersampling (27,732 reports)	14,626	52.74
LLM-predicted labels, dominant topic statistics (27,732 reports)	15,675	56.52

3.4 Training Set Distribution

The training set consists of the 27,732 reports obtained after the cleaning and undersampling described in Section 3.3, labelled in two ways, as mentioned above.

The distribution of the labels under the two label sets is reported in Table 3.3 (main topics) and Table 3.4 (subtopics). The total reported in each table is the total number of assigned labels, which is higher than the number of reports because a report can carry more than one label. On average, the rule-based labels assign more labels per

report than the LLM-predicted labels, both at the main topic and at the subtopic level.

Table 3.3: Distribution of main topics in the training set (27,732 reports), under the rule-based labels and the GPT-4.1 LLM-predicted labels. n is the number of reports containing the label. Percentages are the share of reports that contain it. The total is the number of assigned labels.

	Rule-based		**GPT-4.1**	
Main topic	**n**	**%**	**n**	**%**
opruimen, afval en onderhoud	14626	52.74	15675	56.52
veiligheid en omgeving	11729	42.29	12689	45.76
parkeren	6126	22.09	5202	18.76
vervoer	4947	17.84	2963	10.68
dagelijks leven en sociale gelegenheden	3708	13.37	2016	7.27
maatschappelijke ondersteuning	829	2.99	135	0.49
zorg	597	2.15	255	0.92
wonen en ondernemen	514	1.85	245	0.88
burgerzaken	430	1.55	73	0.26
financiële ondersteuning	106	0.38	33	0.12
Total assigned labels	**43612**		**39286**	

3.5 Test Set

The evaluation uses two test sets: the *full human-validated test set* (1,303 reports) and the *re-annotated test set* (182 reports), which is a subset of the first. The two are built from the same validation process, described below.

The test set was not annotated from scratch. A domain expert at the partner company went through an unsystematic selection of 1,305 reports and, for each one, marked the labels assigned by the rule-based system as correct or incorrect. The expert's role was limited to verifying the existing labels: no new label was added, even when a relevant label was missing. The validation therefore tells whether the rule-based labels are correct, not what the correct labels should be.

Of the 1,305 reviewed reports, 1,121 were marked as correct and 184 as misclassified. Two of the misclassified reports were excluded as explained Section 3.5.1, leaving 182. The evaluation is carried out on the remaining 1,303 reports, which form the *full human-validated test set*: the 1,121 reports whose rule-based labels were confirmed, and the 182 reports that were re-annotated by the author and form the *re-annotated test set*.

The two test sets play different roles. In the full human-validated test set, the large majority of the reports (1,121 of 1,303) are cases that the rule-based system already classifies correctly. This introduces a selection bias towards the rule-based labels. The re-annotated test set contains only the reports the rule-based system got wrong, re-annotated by hand, and is therefore relatively more independent of the rule-based labels, and less affected by this selection bias. The implications of this difference for the comparison of the approaches are discussed in Chapter 6.

The distribution of the two test sets is reported in Table 3.5 (main topics) and Table 3.6 (subtopics). The two distributions differ. some topics (for example *veiligheid en omgeving*) appear with a different frequency in the re-annotated set.

Table 3.4: Distribution of subtopics in the training set (27,732 reports), under the rule-based labels and the GPT-4.1 LLM-predicted labels. n is the number of reports containing the subtopic; percentages are the share of reports that contain it. The last row reports the total number of assigned labels, which exceeds the number of reports because a report can carry more than one label.

	Rule-based		**GPT-4.1**	
Subtopic	**n**	**%**	**n**	**%**
onderhoud omgeving	9476	34.17	7210	26.00
vuil/ongedierte overlast	5738	20.69	4855	17.51
algemene veiligheid	4402	15.87	2875	10.37
parkeeroverlast	3666	13.22	3932	14.18
infrastructuur	3228	11.64	1934	6.97
buurt ambiance	2921	10.53	1005	3.62
overlast personen	2641	9.52	2046	7.38
bruikbaarheid/beschikbaarheid_afvalcontainers	2544	9.17	1104	3.98
bereikbaarheid/beschikbaarheid_parkeerplek	2527	9.11	1058	3.82
ophalen huiscontainers/grofvuil	1707	6.16	2393	8.63
verlichting	1706	6.15	1236	4.46
verkeersmaatregelen	1670	6.02	464	1.67
verkeersveiligheid	1350	4.87	2837	10.23
geluidsoverlast	1348	4.86	1339	4.83
criminaliteit	1262	4.55	283	1.02
geuroverlast	1177	4.24	603	2.17
huisdierenoverlast	844	3.04	1138	4.10
parkeervergunning/abonnement	824	2.97	53	0.19
parkeergarage	682	2.46	102	0.37
verkeersovertreding	518	1.87	619	2.23
hangjongeren	434	1.56	398	1.44
huur en verhuur	415	1.50	244	0.88
hulp aan dak- en thuislozen	360	1.30	253	0.91
evenementen, feestdagen en herdenkingen	333	1.20	398	1.44
hulpmiddelen en woonvoorzieningen	308	1.11	135	0.49
schade en claims	288	1.04	598	2.16
documentaanvraag	203	0.73	13	0.05
parkeerkosten	174	0.63	33	0.12
luchtkwaliteit	131	0.47	195	0.70
burgemeester	122	0.44	25	0.09
verhuizen	106	0.38	42	0.15
vaccinatie	77	0.28	3	0.01
financiële regelingen	61	0.22	15	0.05
overlijden	48	0.17	19	0.07
parkeren bezoekers	42	0.15	18	0.06
parkeerterminal	28	0.10	28	0.10
bijstand	11	0.04	17	0.06
Total assigned labels	**53385**		**39572**	

Table 3.5: Distribution of main topics in the full human-validated test set (1,303 reports) and in the re-annotated test set (182 reports). n is the number of reports containing the main topic; percentages are the share of reports that contain it. The last row reports the total number of assigned labels, which exceeds the number of reports because a report can carry more than one label.

	Full test		**Re-annotated**	
Main topic	**n**	**%**	**n**	**%**
opruimen, afval en onderhoud	799	61.23	100	54.95
veiligheid en omgeving	515	39.46	32	17.58
dagelijks leven en sociale gelegenheden	136	10.42	51	28.02
parkeren	123	9.43	17	9.34
vervoer	99	7.59	8	4.40
zorg	12	0.92	6	3.30
burgerzaken	6	0.46	0	0.00
financiële ondersteuning	2	0.15	0	0.00
wonen en ondernemen	2	0.15	0	0.00
maatschappelijke ondersteuning	2	0.15	0	0.00
Total assigned labels	**1696**		**214**	

3.5.1 Re-Annotations of the Re-Annotated Test set

The 182 reports were re-annotated from scratch by the author. These should represent the more difficult cases, and the rule-based labels attached to them are unreliable.

The annotation was carried out one report at a time. Each report was annotated against the taxonomy of 10 main topics and 37 subtopics, using the official description of each topic (Section 3.2). The reports were read in English translation to overcome the Dutch language barrier. The rule-based labels were deliberately not consulted, so that the original (incorrect) predictions could not influence the new annotation. Each report was then assigned the main topics and subtopics that matched its content.

Two of the 184 misclassified reports were excluded from this annotation because their text was too ambiguous, the translation unreliable, and limited time available to familiarize with the taxonomy. The procedure has a further, more general limitation: since the annotation was carried out by a single annotator, no inter-annotator agreement could be computed. These limitations are discussed in Chapter 6.

As the Tables 3.5 and 3.6, the re-annotated test set covers only a subset of the taxonomy (6 of the 10 main topics and 18 of the 37 subtopics). This is an additional limitation, discussed in Chapter 6.

Table 3.6: Distribution of subtopics in the full human-validated test set (1,303 reports) and in the re-annotated test set (182 reports). n is the number of reports containing the subtopic; percentages are the share of reports that contain it. The last row reports the total number of assigned labels, which exceeds the number of reports because a report can carry more than one label.

	Full test		**Re-annotated**	
Subtopic	**n**	**%**	**n**	**%**
onderhoud omgeving	431	33.03	55	30.22
vuil/ongedierte overlast	355	27.20	6	3.30
criminaliteit	280	21.46	11	6.04
bruikbaarheid/beschikbaarheid_afvalcontainers	163	12.49	0	0.00
ophalen huiscontainers/grofvuil	153	11.72	40	21.98
algemene veiligheid	112	8.58	11	6.04
verlichting	82	6.28	0	0.00
parkeeroverlast	79	6.05	16	8.79
overlast personen	75	5.75	6	3.30
infrastructuur	72	5.52	4	2.20
schade en claims	64	4.90	45	24.73
buurt ambiance	58	4.44	3	1.65
geuroverlast	49	3.75	0	0.00
bereikbaarheid/beschikbaarheid_parkeerplek	49	3.75	0	0.00
verkeersmaatregelen	40	3.07	2	1.10
geluidsoverlast	36	2.76	7	3.85
verkeersveiligheid	25	1.92	2	1.10
parkeergarage	15	1.00	0	0.00
evenementen, feestdagen en herdenkingen	10	0.77	1	0.55
hangjongeren	10	0.77	1	0.55
verkeersovertreding	8	0.61	2	1.10
parkeervergunning/abonnement	7	0.54	0	0.00
huisdierenoverlast	6	0.46	0	0.00
hulp aan dak- en thuislozen	6	0.46	1	0.55
verhuizen	4	0.31	0	0.00
burgemeester	3	0.23	0	0.00
huur en verhuur	3	0.15	0	0.00
parkeerkosten	2	0.15	0	0.00
hulpmiddelen en woonvoorzieningen	2	0.15	0	0.00
parkeren bezoekers	2	0.15	1	0.55
bijstand	1	0.08	0	0.00
documentaanvraag	1	0.08	0	0.00
financiële regelingen	1	0.08	0	0.00
vaccinatie	1	0.08	0	0.00
luchtkwaliteit	1	0.08	0	0.00
parkeerterminal	1	0.08	0	0.00
overlijden	1	0.08	0	0.00
Total assigned labels	**2195**		**213**	

Chapter 4

Methodology

4.1 Multi-Label Topic Classification

The goal of this thesis is to identify the best performing classification approach for Multi-label topic classification (MLTC) in the governance domain. As discussed so far, one of the main challenges of this study is the scarcity of annotated labels. This chapter describes the techniques used to address this problem and the reasons behind their selection, as well as the approaches used to perform MLTC on Dutch citizens' reports.

The methodology follows this design. No human-annotated labels are available for the training data: the reports carry only the labels produced by the company's rule-based system and are not human-validated. For this reason, a generative language model is used to automatically generate the training labels (LLM-predicted labels). To decide the approach, two generative language models are first evaluated through zero-shot prediction on the *full human-validated test set* and the *re-annotated test set* (see Section 3.5), comparing a one-step and a two-step approach (Section 4.1.3). The best performing classification approach on both test sets is the one selected to generate LLM-predicted labels for training. The resulting LLM-predicted labels are then used to fine-tune an encoder model (BERTje, (de Vries et al., 2019)). The BERTje model is evaluated on the same two test sets. The performance is then compared to the zero-shot prediction of the generative language model and an encoder fine-tuned on the rule-based labels produced by the partner company's system. This last model serves as a reference for the labels that the company has already available, against which the quality of the LLM-predicted labels can be assessed.

The chapter is structured as follows. First, generative language models and their zero-shot capabilities are introduced, together with the prompting strategy used for LLM-predicted labels generation and zero-shot prediction (Subsection 4.1.1. Then, transfer learning and the BERTje architecture are described (Devlin et al., 2018; de Vries et al., 2019) in Subsection 4.1.2. The one-step and two-step classification approaches and the problem transformation are presented next in Subsection 4.1.3. Then, the three experiments of this work are detailed (Section 4.2). Finally, the evaluation metrics are described in Section 4.3.

4.1.1 Generative Language Models for Zero-Shot Prediction

Generative language models are autoregressive, decoder-only models. They generate text by relying on the preceding tokens (Vaswani et al., 2017). Unlike encoder models, they lack bidirectionality, since they are trained to predict one token after another. Models such as GPT and Llama use only the decoder component, whose self-attention is restricted to the preceding tokens (Alammar, 2018). In this work, two such models are used: GPT-4.1 (OpenAI et al., 2024) and Llama 3.3 70B Instruct (Grattafiori et al., 2024).

These models are used in two ways:

- **Zero-shot prediction.** The models perform zero-shot prediction of main topics and subtopics on both test sets, in both a one-step and a two-step approach (Section 4.1.3), in order to evaluate their performance as multi-label classifiers.
- **LLM-predicted labels generation.** The best performing model on the two test sets is used to generate the LLM-predicted labels for the entire training set, which are then used to fine-tune a BERTje model.

GPT-4.1 with the one-step classification approach is the one used to generate these labels. This choice is based on the comparison of the configurations reported in Chapter 5 (Table 5.1).

Zero-shot Prompting

Different prompting techniques can be used to make a generative language model perform a task it was not explicitly trained for, a task the model is unlikely to have seen during training (Brown et al., 2020). Few-shot prompting, a form of in-context learning, provides the model with several examples of the task, potentially filling the context window (Brown et al., 2020). One-shot prompting provides a single example. This work focuses on zero-shot prompting, the most straightforward way to make a generative language model perform a task without any fine-tuning: the model receives only the instructions, and its parameters are not updated (Naim et al., 2025).

Kong et al. (2024) outline the trade-offs between prompting and fine-tuning. While fine-tuning requires substantial computational resources, prompting bypasses this requirement but does not alter the underlying model representations. The performance of a model instructed through prompting therefore depends strictly on the capabilities it acquired during pre-training. Moreover, Brown et al. (2020) note that zero-shot evaluation poses an additional challenge, since the model has to interpret the task without any demonstration, and the instructions can easily become ambiguous if they are not precisely defined.

Because the label space is hierarchical, a generative language model can be prompted to classify a report in one-step or in two-steps. The two classification approaches are described in Section 4.1.3.

Both approaches are evaluated for both models on the two test sets. The corresponding prompts are reported below. The two classification approaches are compared in Chapter 5 (Table 5.1). This comparison motivates the approach used to generate the LLM-predicted labels.

Both prompts instruct the model to assign main topics and subtopics to each citizens' report and to return the output in a structured JSON format.

Listing 4.1: Prompt used for zero-shot prediction and LLM-predicted labels generation with a one-step approach.

```
"""You are a Dutch expert annotator for citizens' reports (such as
    complaints, requests, general inquiries).

Your task: given a Dutch text, assign the appropriate main topics and
    subtopics.

RULES:

- Only use labels from the lists below. Do not invent new labels.
- A text can have multiple main topics.
- Each main topic can have one or more subtopics.
- For each report, there are up to a maximum of 3 main topics and 3
    subtopics.

MAIN TOPICS AND THEIR SUBTOPICS:
{json.dumps(LABEL_MAP, indent=2)}

RESPONSE FORMAT JSON ONLY:
{{
"labels": [
{{"topic": "main_topic_name", "subtopics": ["subtopic_1", "subtopic_2"]}},
{{"topic": "another_topic", "subtopics": ["subtopic_3"]}}
]
}}

If no topics apply, respond with:
{{"topic": "No Topic Found", "subtopics": ["No Subtopic Found"]}},
"""
```

Listing 4.2: Prompt used for zero-shot prediction with a two-step approach.

```
SYSTEM_PROMPT_MAIN = """You are a Dutch expert annotator for citizens'
    reports (such as complaints, requests, general inquiries).

Your task: given a Dutch text, assign the main topics that genuinely apply
     to it.

RULES:

- Only use main topics from the list below. Do not invent new labels.
- Assign a main topic only if the text genuinely concerns it. Do not add
    topics to reach a certain number.
- A text may concern a single main topic, or several. Assign only one if
    only one applies.
- List each applicable main topic at most once.

MAIN TOPICS:
{json.dumps(MAIN_TOPICS, indent=2)}

RESPONSE FORMAT JSON ONLY:
{{
"topics": ["main_topic_name"]
}}
```

```
If no topic applies, respond with:
{{"topics": ["No Topic Found"]}}
"""

SYSTEM_PROMPT_SUB = """You are a Dutch expert annotator for citizens'
    reports.

Your task: given a Dutch text and a single main topic already assigned to
    it, choose the subtopics that apply.

RULES:

- Only choose subtopics from the restricted list provided in the user
    message. Do not invent new labels.
- You may choose one or more subtopics, up to a maximum of 3.

RESPONSE FORMAT JSON ONLY:
{"subtopics": ["subtopic_1", "subtopic_2"]}

If no subtopic applies, respond with:
{"subtopics": ["No Subtopic Found"]}
"""
```

4.1.2 Transfer Learning and BERTje

Supervised learning typically requires a considerable amount of annotated data for the target task. A standard way to overcome this limitation is *transfer learning*, a paradigm that tackles the scarcity of annotated data by transferring knowledge from a source domain to a target domain (Pan and Yang, 2010). This makes it possible to train a classifier for the governance domain without a large manually annotated dataset. The model reuses the linguistic knowledge acquired during pre-training and adapts it to the target task.

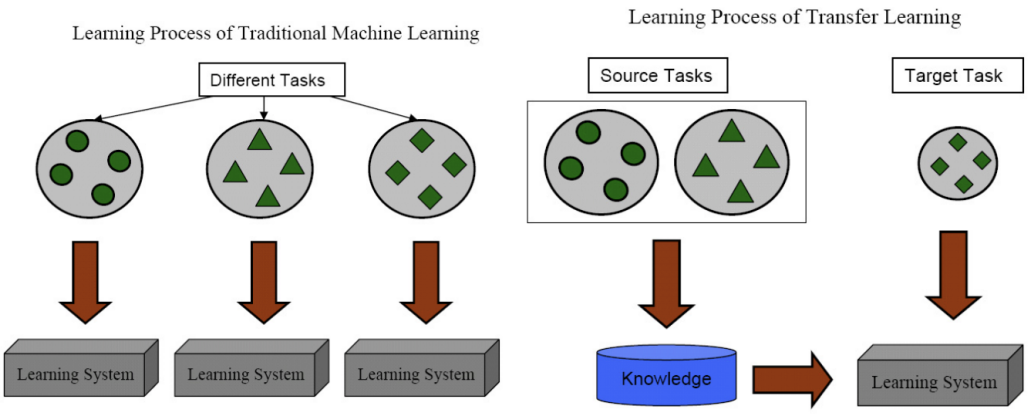


Figure 4.1: Visualization of transfer learning (Pan and Yang, 2010).

BERT is an example of the transfer learning paradigm and is built on the Transformer architecture introduced by Vaswani et al. (2017). The original Transformer follows an encoder-decoder structure: the encoder builds a representation of the input sequence, while the decoder generates the output tokens. The two components can be used independently according to the task: the encoder is used for tasks requiring text comprehension, and the decoder for text generation. Following this logic, BERT and BERTje are based exclusively on the encoder, while generative models such as GPT use the decoder (Section 4.1.1).

The BERT framework (Devlin et al., 2018) involves two phases: pre-training and fine-tuning. In the first phase, the model is pre-trained in a self-supervised manner on a massive corpus, learning broad linguistic representations. In the second phase, the pre-trained model is fine-tuned on a specific target task: by adapting the final output layer, the linguistic knowledge acquired during pre-training is leveraged, and the model's weights are updated for the target task and domain, in this case MLTC for Dutch citizens' reports in the governance domain. The model used in this study is BERTje (de Vries et al., 2019), a BERT model pre-trained on a large collection of Dutch corpora, which makes it suited to processing the Dutch reports without machine translation.

4.1.3 One-Step and Two-Step Classification Approach

As described in Chapter 3, the labels are organized hierarchically: each main topic is associated with one or more subtopics. A generative language model can process this structure directly, following the taxonomy specified in the prompt. An encoder-only model cannot: the multi-label problem must first be reformulated into a format it can process.

Given the hierarchical structure of the labels, two strategies are possible: one-step and two-step classification. The meaning of the one-step approach differs between encoder and decoder models. For an encoder such as BERTje, a one-step formulation would flatten the main topics and their corresponding subtopics into a single label space, since the model outputs a vector of independent binary decisions and does not reproduce the hierarchy between main topics and subtopics. For a generative language model, instead, the one-step prompt returns the full hierarchical structure (each main topic with its subtopics) in a single JSON output (see the prompt in Listings 4.1 and 4.2). Here the one-step formulation does not discard the hierarchy.

The two formulations differ in more than the number of API calls. In the two-step approach, the subtopics are predicted in a second step that is dependent on the main topics predicted in the first step: in the second step the subtopics are predicted only for the reports assigned to each predicted main topic, so an error in the main topic prediction propagates to the subtopic prediction. In the one-step approach this dependency is absent, since main topics and subtopics are predicted jointly. For BERTje, the two-step approach is adopted, since it preserves the hierarchical structure that the one-step classification approach would discard. Consequently, the model later referred to as the $\text{BERTje}_{\text{LLM}}$ is a two-step classifier trained on labels that GPT-4.1 produced in a one-step classification approach: for the generative language model, one-step refers to how the training labels were generated, not to how BERTje classifies. Moreover, Szabo (2024) found the two approaches to produce comparable results in the same domain, which is why a one-step BERTje is not evaluated separately in this work. The generative language models do not require this transformation, but they can still be prompted

in either way. In this project, they are tested in both classification approaches, and the one-step approach is the one used to generate the LLM-predicted labels for training (Section 4.1.1).

The strategy for handling multi-label classification adopted in this work is *Binary Relevance*. Transforming a multi-label problem into a set of binary ones is among the most straightforward ways to tackle the task (Herrera et al., 2016; Tsoumakas and Katakis, 2007). Binary Relevance, also referred to as one-vs-all (OVA) or one-vs-the-rest (OVR), requires training an independent binary classifier for every label, and therefore assumes label independence, ignoring any potential correlation between labels. For example, if the main topics are *Sport* and *Politics*, two binary classifiers are trained, each deciding whether a text belongs to that topic or not.

In practice, BERTje does not require a separate classifier per label, since a single model produces a vector of independent binary decisions in parallel. In the first step, a single BERTje model produces a vector of binary decisions, one for each of the ten main topics. In the second step, ten further BERTje models are trained, one per main topic, each with a classification head whose output dimension equals the number of subtopics of that main topic.

4.2 Experiments

This section describes the three experiments of this work and the parameters used in each. The training set is the same across all BERTje experiments: it results from the undersampling procedure described in Chapter 3, and it is identical in Experiment 1 and Experiment 3. For all BERTje models, the hyperparameters follow the configuration recommended for BERT fine-tuning (Devlin et al., 2018) and are reported in Table 4.1; because of time and computational constraints, no hyperparameter tuning was performed. The data are split into a training set and a test set only and no separate validation set is used. The fine-tuning was executed on a 10th Gen Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz with 16GB of RAM. A fixed random seed of 42 is used for all BERTje experiments.

Experiment 1 — BERTje fine-tuned on rule-based labels

Experiment 1 measures the performance of a BERTje model fine-tuned on the rule-based labels (hereafter, $\text{BERTje}_{\text{RB}}$) produced by the partner company's system (Chapter 3). The model is BERTje (`GroNLP/bert-base-dutch-cased`), introduced in Section 4.1.2. It is fine-tuned to perform two-step classification, as described in Section 4.1.3, and evaluated separately on each of the two test sets. The classification head is an output layer with one unit per label and a sigmoid activation on each unit. Binary Cross-Entropy is used as the loss function, consistently with the multi-label formulation, in which each label is treated as an independent binary decision.

This model cannot be considered a lower-bound baseline. Because of the selection bias of the full human-validated test set described in Chapter 3, a model fine-tuned on the rule-based labels is advantaged by the way that test set is constructed.

Experiment 2 — Zero-shot LLM prediction

Experiment 2 compares GPT-4.1 and Llama 3.3 70B Instruct on the zero-shot prediction on both test sets. Only the texts are given to the models and the corresponding

Hyperparameter	Value
Max sequence length	128
Batch size	8
Epochs	3
Learning rate	2e-5
Threshold	0.5
Optimizer	AdamW
Loss function	Binary Cross-Entropy
Random seed	42

Table 4.1: Hyperparameters used for BERTje fine-tuning. The configuration follows the recommendations in (Devlin et al., 2018).

labels are withheld, so that the models rely solely on the input reports. Each model returns its predictions in JSON format, which are then converted into binary vectors and evaluated against both test sets. The models are accessed through Microsoft Azure[1]. The only parameter modified is the *temperature*, set to 0.3: it is kept low to reduce the variability of the predictions and make the output more deterministic, while the remaining parameters are left at their default values.

Experiment 2a — One-step approach

In the one-step approach, each model predicts all applicable main topics and their subtopics in a single pass, using the prompt in Listing 4.1.

Experiment 2b — Two-step approach

In the two-step approach, each model first predicts the main topics and then, for each predicted main topic, its subtopics, using the prompts in Listing 4.2.

Experiment 3 — BERTje fine-tuned on LLM-predicted labels

Experiment 3 fine-tunes BERTje on the LLM-predicted labels generated by GPT-4.1 with the one-step approach ($\text{BERTje}_{\text{LLM}}$). The training reports, the two-step classification approach, the classification head, and the hyperparameters are identical to those of Experiment 1; the only difference is the label set. This design makes it possible to attribute any difference in performance to the label sets rather than to the model configuration. The model is evaluated on the same two test sets. The aim is to assess whether fully automatically generated LLM-predicted labels can train an encoder to match the $\text{BERTje}_{\text{RB}}$ on the full human-validated test set, despite the bias, and to outperform it on the re-annotated test set.

These three experiments, taken together, make it possible to compare the zero-shot prediction of a generative language model with an encoder fine-tuned on automatically generated LLM-predicted labels, and to evaluate both against a BERTje fine-tuned with rule-based labels.

[1] Microsoft Azure AI Foundry

4.3 Evaluation

All experiments are evaluated separately on both test sets, so that the performance can be assessed independently. The evaluation is carried out on a per-label basis: for each main topic and each subtopic, the predicted labels are compared to the labels in the test set, and Precision (P), Recall (R), and F1-score (F1) are computed for that label. From the per-label results, aggregate scores are then obtained through micro- and macro-averaging. Given the class imbalance of the data, micro-averaged F1 is used as the main metric for comparing the approaches (as thoroughly explained in Section 5.1); the full set of per-label results and the different averaging methods are reported in Chapter 5. The results of the experiments are presented in the following chapter.

Chapter 5

Results

This chapter presents the results of the three experiments. First, Section 5.1 illustrates the reason for focusing on micro-F1 rather than another averaging method. Then, Section 5.2 gives an overview of the results: first the zero-shot prediction of the two generative language models (Subsection 5.2.1), then the comparison of the encoder approaches (Subsection 5.2.2. Section 5.3 reports the per-topic results of the preferred model, BERTje$_{\text{LLM}}$ (fine-tuned on GPT-4.1 LLM-predicted labels). Section 5.4 presents the error analysis. The evaluation metrics are defined in Section 4.3.

5.1 Averaging Methods

The tables report two averaging methods for precision, recall and F1-score. Macro average is the unweighted mean of the per-topic scores, so every topic counts equally regardless of its support. This is useful to show how the model performs on the under-represented topics. Micro average instead aggregates the true positives, false positives and false negatives across all topics, so each topic contributes in proportion to its support. Because of class imbalance, where many topics have only one or two instances and are often missed entirely, micro-F1 is used as the main metric for comparing the approaches (Charte et al., 2015).

5.2 Overall Results

5.2.1 Results: Zero-Shot Prediction (GPT-4.1 & Llama 3.3 50B Instruct)

This section compares the two generative language models, GPT-4.1 and Llama 3.3 70B Instruct, as zero-shot multi-label classifiers, in both the one-step and the two-step classification approach (see Section 4.1.3). Table 5.1 reports their micro- and macro-averaged results on main topics and subtopics, on the two test sets. GPT-4.1 outperforms Llama 3.3 70B Instruct across both approaches and both test sets. GPT-4.1 one-step obtains the highest micro-F1 on main topics on both test sets and is competitive on subtopics, reaching a micro-F1 of 0.48 on the full human-validated test set, the same as Llama 3.3 70B Instruct two-step. It was therefore selected to generate the LLM-predicted labels, also because one-step requires a single API call per report rather than two. The two-step approach scores marginally higher on subtopics on the re-annotated test set (micro-F1 0.42 for GPT-4.1 two-step, 0.04 higher than one-step).

Table 5.1: Zero-shot LLM classification: one-step vs two-step results for GPT-4.1 and Llama 3.3 70B Instruct, on main topics and subtopics, full and re-annotated test sets. Bold marks the best micro-F1 per column. GPT-4.1 one-step is the best-performing model (or at least comparable) in most cases and is selected for generating the GPT-4.1 LLM-predicted labels.

			Micro avg			Macro avg		
Model	**Approach**	**Test set**	**P**	**R**	**F1**	**P**	**R**	**F1**
Main topics								
GPT-4.1	one-step	Full	0.79	0.75	**0.77**	0.58	0.45	0.49
GPT-4.1	one-step	Re-annotated	0.63	0.68	**0.65**	0.56	0.56	0.51
GPT-4.1	two-step	Full	0.84	0.66	0.74	0.38	0.34	0.29
GPT-4.1	two-step	Re-annotated	0.68	0.60	0.64	0.53	0.46	0.44
Llama 3.3 70B	one-step	Full	0.49	0.53	0.51	0.16	0.16	0.16
Llama 3.3 70B	one-step	Re-annotated	0.55	0.65	0.60	0.50	0.52	0.46
Llama 3.3 70B	two-step	Full	0.72	0.74	0.73	0.29	0.38	0.30
Llama 3.3 70B	two-step	Re-annotated	0.56	0.66	0.61	0.37	0.51	0.41
Subtopics								
GPT-4.1	one-step	Full	0.52	0.45	**0.48**	0.47	0.45	0.40
GPT-4.1	one-step	Re-annotated	0.38	0.38	0.38	0.50	0.56	0.45
GPT-4.1	two-step	Full	0.53	0.41	0.47	0.32	0.28	0.26
GPT-4.1	two-step	Re-annotated	0.46	0.39	**0.42**	0.40	0.44	0.38
Llama 3.3 70B	one-step	Full	0.17	0.22	0.19	0.08	0.09	0.07
Llama 3.3 70B	one-step	Re-annotated	0.27	0.41	0.33	0.32	0.47	0.29
Llama 3.3 70B	two-step	Full	0.44	0.52	**0.48**	0.32	0.39	0.30
Llama 3.3 70B	two-step	Re-annotated	0.32	0.48	0.38	0.32	0.50	0.33

5.2.2 Results: Encoder Approaches (BERTje)

This section compares the $\text{BERTje}_{\text{RB}}$ and $\text{BERTje}_{\text{LLM}}$ on the two test sets, using the summary results in Table 5.2 (main topics) and Table 5.3 (subtopics). On the full human-validated test set, $\text{BERTje}_{\text{RB}}$ obtains the highest micro-F1, both on main topics (0.86) and subtopics (0.77). However, this result is not a sign of a better model. As described in Section 3.5, most of the full human-validated test set consists of reports rule-based system classifies correctly, so a model trained on those same labels is favored by the way the test set is built. The re-annotated test set shows this more clearly: on this set, $\text{BERTje}_{\text{RB}}$ drops from 0.86 to 0.57 on main topics, a fall of 0.29 points. The other two approaches fall much less on the same set (GPT-4.1 from 0.77 to 0.65, $\text{BERTje}_{\text{LLM}}$ from 0.78 to 0.64, that is 0.12 and 0.14 points respectively). This larger drop indicates that the $\text{BERTje}_{\text{RB}}$ has learned the patterns of the rule-based system, and does not generalize to the cases the rule-based system misclassifies. For this reason $\text{BERTje}_{\text{RB}}$ is set aside for the remainder of the study. Setting $\text{BERTje}_{\text{RB}}$ aside does not change the comparison between the remaining approaches.

GPT-4.1 zero-shot prediction and $\text{BERTje}_{\text{LLM}}$ perform comparably. On the full human-validated test set they are almost equivalent on the main topics (0.78 for $\text{BERTje}_{\text{LLM}}$, 0.77 for GPT-4.1) and on the subtopics (0.46 and 0.48). The same is true for the re-annotated test set (0.64 and 0.65 on main topics, 0.38 for both on subtopics). An encoder trained only on labels generated by GPT-4.1 therefore reaches the same level as GPT-4.1 itself, including on the re-annotated test set. This observation and its implications are discussed in Chapter 6.

Table 5.2: Summary of main topic classification results. Bold marks the best micro-F1 per test set.

		Micro avg			Macro avg		
Model	**Test set**	**P**	**R**	**F1**	**P**	**R**	**F1**
Zero-shot LLM (one-step)							
GPT-4.1	Full	0.79	0.75	0.77	0.58	0.45	0.49
GPT-4.1	Re-annotated	0.63	0.68	**0.65**	0.56	0.56	0.51
BERTje							
BERTje$_{RB}$	Full	0.82	0.90	**0.86**	0.54	0.84	0.60
BERTje$_{RB}$	Re-annotated	0.52	0.64	0.57	0.45	0.61	0.46
BERTje$_{LLM}$	Full	0.78	0.78	0.78	0.62	0.60	0.57
BERTje$_{LLM}$	Re-annotated	0.60	0.68	0.64	0.47	0.52	0.45

Table 5.3: Summary of subtopic classification results. Bold marks the best micro-F1 per test set.

		Micro avg			Macro avg		
Model	**Test set**	**P**	**R**	**F1**	**P**	**R**	**F1**
Zero-shot LLM (one-step)							
GPT-4.1	Full	0.52	0.45	0.48	0.47	0.45	0.40
GPT-4.1	Re-annotated	0.38	0.38	**0.38**	0.50	0.56	0.45
BERTje							
BERTje$_{RB}$	Full	0.70	0.85	**0.77**	0.51	0.77	0.57
BERTje$_{RB}$	Re-annotated	0.30	0.38	0.34	0.34	0.54	0.36
BERTje$_{LLM}$	Full	0.49	0.43	0.46	0.36	0.41	0.33
BERTje$_{LLM}$	Re-annotated	0.38	0.38	**0.38**	0.41	0.49	0.39

A consistent gap is present between micro- and macro-averaged F1 across all approaches, which indicates that performance is driven by the frequent topics and is weak on the rare ones (more detailed per-topic results to be found in Appendix A). As expected, at the subtopic level all approaches reach considerably lower results than at the main topic level, confirming that the more fine-grained part of the taxonomy is the harder one. The highest micro-F1 among the compared approaches for subtopics is 0.48 (GPT-4.1 zero-shot prediction), once BERTje$_{RB}$ is set aside for the bias discussed above. The error analysis in Section 5.4 therefore focuses on BERTje$_{LLM}$, the preferred model for the operational reasons discussed in Chapter 6.

The full results of the other models and approaches and the confusion matrices are included in Appendix A and Appendix B.

5.3 Detailed results: BERTje$_{LLM}$

This section reports the per-topic results of BERTje$_{LLM}$, on the full human-validated test set and on the re-annotated test set. The full results of the other models and approaches are reported in Appendix A.

5.3.1 Per-topic results on the full human-validated test set

Table 5.4: $\text{BERTje}_{\text{LLM}}$ – main topics, full human-validated test set.

topic	P	R	F1	Supp.
veiligheid en omgeving	0.77	0.71	0.74	515
opruimen, afval en onderhoud	0.84	0.95	0.89	799
dagelijks leven en sociale geleg.	0.38	0.29	0.33	136
parkeren	0.76	0.79	0.77	123
vervoer	0.69	0.54	0.60	99
zorg	0.80	0.33	0.47	12
financiële ondersteuning	0.00	0.00	0.00	2
wonen en ondernemen	0.29	1.00	0.44	2
burgerzaken	0.71	0.83	0.77	6
maatschappelijke ondersteuning	1.00	0.50	0.67	2
micro avg	0.78	0.78	0.78	1696
macro avg	0.62	0.60	0.57	1696

On main topics, $\text{BERTje}_{\text{LLM}}$ reaches a micro-F1 of 0.78, with precision and recall in balance (0.78 for both). The highest scores are on *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*) (0.89), which is also the most frequent topic, and on *parkeren* (*parking*) (0.77). The other frequent topic, *veiligheid en omgeving* (*safety and environment*) (support 515), scores 0.74. One topic, *financiële ondersteuning* (*financial support*), scores 0.00, but it has a support of only 2 reports; other equally rare topics are still predicted, such as *burgerzaken* (*civil affairs*) and *maatschappelijke ondersteuning* (*social support*). The clearest weak point with a relevant support is *dagelijks leven en sociale gelegenheden* (*daily life and social occasions*), which scores 0.33 over 136 reports.

Table 5.5: $\text{BERTje}_{\text{LLM}}$ – subtopics, full human-validated test set.

topic	P	R	F1	Supp.
criminaliteit	0.78	0.03	0.05	280
algemene veiligheid	0.33	0.47	0.39	112
verlichting	0.83	0.88	0.85	82
overlast personen	0.22	0.17	0.20	75
geluidsoverlast	0.27	0.94	0.42	36
geuroverlast	0.25	0.02	0.04	49
vuil/ongedierte overlast	0.58	0.59	0.58	355
onderhoud omgeving	0.68	0.49	0.57	431
bruikb./beschikb. afvalcontainers	0.46	0.44	0.45	163
huisdierenoverlast	0.24	0.83	0.37	6
buurt ambiance	0.24	0.14	0.18	58
evenementen, feestd. en herd.	0.04	1.00	0.07	1
hangjongeren	0.75	0.90	0.82	10
ophalen huiscont./grofvuil	0.44	0.58	0.50	153
parkeergarage	0.80	0.31	0.44	15
verkeersmaatregelen	0.33	0.07	0.12	40

topic	P	R	F1	Supp.
infrastructuur	0.58	0.54	0.56	72
bereikb./beschikb. parkeerplek	0.28	0.24	0.26	49
parkeeroverlast	0.64	0.73	0.69	79
verkeersveiligheid	0.21	0.64	0.31	25
verkeersovertreding	0.10	0.12	0.11	8
schade en claims	0.45	0.31	0.37	64
burgemeester	0.67	0.67	0.67	3
hulp aan dak- en thuislozen	0.83	0.83	0.83	6
bijstand	0.00	0.00	0.00	1
huur en verhuur	0.29	1.00	0.44	3
verhuizen	0.75	0.75	0.75	4
parkeervergunning/abonnement	0.00	0.00	0.00	7
parkeerkosten	0.00	0.00	0.00	2
documentaanvraag	0.00	0.00	0.00	1
hulpmiddelen en woonvoorz.	1.00	0.50	0.67	2
financiële regelingen	0.00	0.00	0.00	1
vaccinatie	0.00	0.00	0.00	1
luchtkwaliteit	0.00	0.00	0.00	1
parkeerterminal	0.00	0.00	0.00	1
parkeren bezoekers	0.00	0.00	0.00	2
overlijden	0.33	1.00	0.50	1
micro avg	0.49	0.43	0.46	2195
macro avg	0.36	0.41	0.33	2195

Many subtopics have a support of only one or two reports. Their per-topic scores (including the 0.00 values) are reported for completeness but are not reliable, since a single prediction changes the score entirely. The same caution applies to the re-annotated test set, given its size (182 reports).

On subtopics, the micro-F1 drops to 0.46.A notable case is *criminaliteit* (*criminality*), which has a high precision (0.78) but a very low recall (0.03) over a support of 280: the model almost never assigns this label. This is the same under-prediction that GPT-4.1 shows on the LLM-predicted labels used for training (Table 3.4). The case is different for *geluidsoverlast* (*noise nuisance*), which is frequent in the training set (1,339 reports) and reaches a recall of 0.94 against a precision of 0.27 (it is over-predicted). In total, 9 subtopics score 0.00, all with low support (between 1 and 7 reports).

5.3.2 Per-topic results on the re-annotated test set

Table 5.6: $\text{BERTje}_{\text{LLM}}$ – main topics, re-annotated test set.

topic	P	R	F1	Supp.
veiligheid en omgeving	0.39	0.88	0.54	32
opruimen, afval en onderhoud	0.72	0.94	0.82	100
dagelijks leven en sociale geleg.	0.62	0.16	0.25	51
parkeren	0.65	0.65	0.65	17
vervoer	0.44	0.50	0.47	8
zorg	0.00	0.00	0.00	6
micro avg	0.60	0.68	0.64	214
macro avg	0.47	0.52	0.45	214

Table 5.7: $\text{BERTje}_{\text{LLM}}$ – subtopics, re-annotated test set.

topic	P	R	F1	Supp.
criminaliteit	0.00	0.00	0.00	11
algemene veiligheid	0.31	0.82	0.45	11
overlast personen	0.30	0.50	0.38	6
geluidsoverlast	0.56	0.71	0.62	7
vuil/ongedierte overlast	0.10	0.83	0.18	6
onderhoud omgeving	0.55	0.43	0.48	54
buurt ambiance	0.67	0.67	0.67	3
evenementen, feestd. en herd.	0.50	1.00	0.67	1
hangjongeren	1.00	1.00	1.00	1
ophalen huiscont./grofvuil	0.67	0.40	0.50	40
verkeersmaatregelen	0.00	0.00	0.00	2
infrastructuur	0.14	0.25	0.18	4
parkeeroverlast	0.77	0.62	0.69	16
verkeersveiligheid	0.07	0.50	0.12	2
verkeersovertreding	0.00	0.00	0.00	2
schade en claims	0.67	0.09	0.16	45
hulp aan dak- en thuislozen	1.00	1.00	1.00	1
parkeren bezoekers	0.00	0.00	0.00	1
micro avg	0.38	0.38	0.38	213
macro avg	0.41	0.49	0.39	213

On the re-annotated test set, the main topic micro-F1 is 0.64, lower than the 0.78 on the full human-validated test set, which might reflect the harder instances. At the subtopic level the micro-F1 is 0.38, a drop of 0.08 points from the full human-validated test set. The macro-F1 moves in the opposite direction at the subtopic level, rising from 0.33 to 0.39. Four subtopics score 0.00 (*criminaliteit*, *verkeersmaatregelen*, *verkeersovertreding* and *parkeren bezoekers*). Except for *criminaliteit*, the others are cases with very low support. As already noted (3.5.1), the re-annotated test set covers fewer main topics and subtopics than the full human-validated test set, so the two sets are not directly comparable.

5.4 Error Analysis

This error analysis focuses on BERTje$_{\text{LLM}}$ (fine-tuned on labels generated by GPT-4.1), the preferred model, and is computed on the full human-validated test set. Its goal is to identify which topics the model confuses with which others, and to look for explanations for these confusions. The analysis is divided into a quantitative part, which measures confusion at the main topic and subtopic levels, and a qualitative part, which groups recurrent error patterns and discusses their possible causes.

The confusion matrices for the main topics of the full human-validated test set are reported in Table 5.1. The subtopic confusion matrices are reported in Appendix B, as their size (37 labels) makes them less practical to read inline. The confusion matrices for the main topic of the re-annotated test set can be found in Figure 5.2, while the subtopic are also in Appendix B for the same reason. Because of the binary relevance transformation, each matrix represents every single topic as a binary classification (topic present vs. topic absent). The analysis focuses on the most relevant classes in the test set according to their support. The five most frequent main topics account for 1,672 of the 1,696 main topic labels in the test set (98.6%). The remaining main topics together account for 24 labels (1.4%), with support ranging from 2 to 12, which is too low to interpret their confusion. The analysis below discusses the five most frequent main topics. All topics are reported in Table 5.8. To ensure the confidentiality of the data, the following examples are fabricated.

5.4.1 Quantitative Analysis for full human-validated test set

Analysis of Main Topic Predictions

Table 5.8: Errors per main topic on the full human-validated test set: false positives (FP) and false negatives (FN). Error rate is computed as (FP + FN) / (support + FP).

Main topic	FP	FN	Error rate
dagelijks leven en sociale gelegenheden	64	96	80.0%
vervoer	24	46	56.9%
veiligheid en omgeving	109	147	41.0%
parkeren	31	26	37.0%
opruimen, afval en onderhoud	148	39	19.7%

In this section, a label is "confused with" another when the model predicts it instead of the correct one. If the model over-predicts a label (high FP), we look at the cases where it was wrongly assigned. If the model frequently misses a label (high FN), we look at which labels were predicted instead. Because reports can have multiple labels at the same time, a single document can cause more than one mistake. For this reason, the total count of these confusions can exceed the total number of False Positives or False Negatives.

Opruimen, afval en onderhoud (cleanup, waste, and maintenance): This topic has 799 positive instances in the test set. The model made 187 errors, 148 FP and 39 FN. It is rarely missed: when it is, it is confused with *veiligheid en omgeving* (*safety and environment*) (34 times), *vervoer* (*transport*) (10 times) and *parkeren* (*parking*) (7

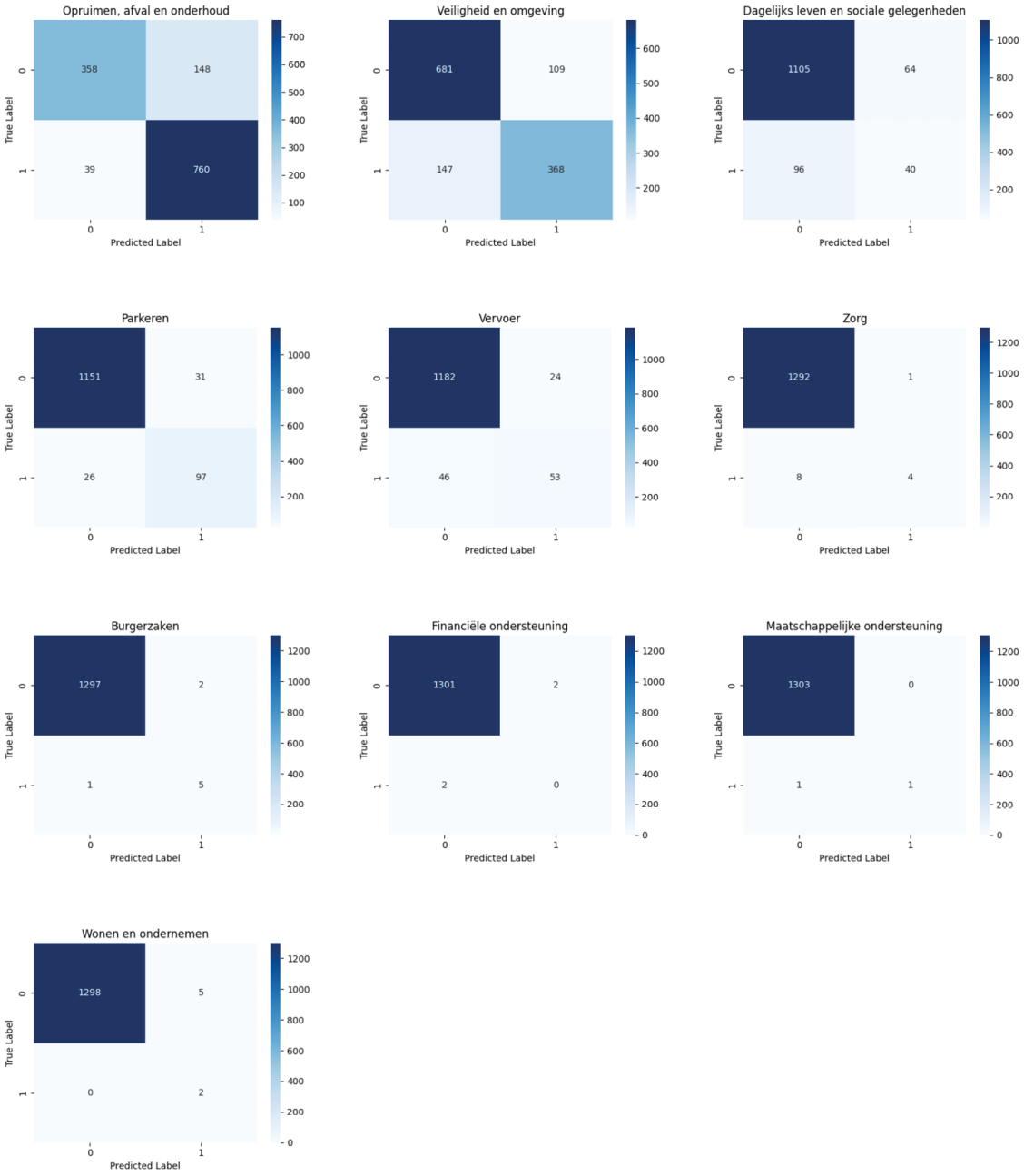


Figure 5.1: Confusion matrices of main topics for BERTje$_{\text{LLM}}$ (full test set

times). The high number of FP shows the opposite problem, namely that the model assigns this topic also to reports where it is not a test set label, as the paragraphs below show.

> "The bike path tiles are sinking and causing dangerous ruts." — predicted as *vervoer* (*transport*).

> "The anti-parking pole is broken off and attached to the light post." — predicted as *parkeren* (*parking*).

Veiligheid en omgeving (Safety and Environment): This topic has 515 positive instances. The model made 256 errors, 109 FP and 147 FN. It is most often confused with *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*) (139 times), then *dagelijks leven en sociale gelegenheden* (*daily life and social events*) (16 times) and *parkeren* (*parking*) (5 times). When a report mentions a concrete element such as waste or damage together with a safety concern, the model tends to assign only the waste topic.

> "At the playground there is glass and rubbish everywhere, children can cut themselves." — predicted as *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*).

> "Damage to trash cans and lamppost caused by fireworks." — predicted as *dagelijks leven en sociale gelegenheden* (*daily life and social events*).

> "Because of a blown-up trash can, the street is littered with mess. I sweep it into the parking bays." — predicted as *parkeren* (*parking*).

Dagelijks leven en sociale gelegenheden (daily life and social events): This topic has 136 positive instances. The model made 160 errors, 64 FP and 96 FN. It is most often confused with *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*) (73 times) and *veiligheid en omgeving* (*safety and environment*) (42 times). With 96 FN over 136 instances, the topic is missed more often than it is found, again in favour of the two dominant topics, as shown in the two examples below.

> "The neighbourhood is declining, there is more and more mess and it feels neglected." — predicted as *opruimen, afval en onderhoud*.

> "The neighbour has placed a camera high up, directly violating our privacy by looking right into our garden." — predicted as *veiligheid en omgeving* (*safety and environment*).

Parkeren (Parking): This topic has 123 positive instances. The model made 57 errors, 31 FP and 26 FN, the lowest error rate among the frequent topics. It is confused with *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*) (16 times), *veiligheid en omgeving* (*safety and environment*) (9 times) and *vervoer* (*transport*) (5 times). Parking reports use a more specific vocabulary, which makes them easier to separate.

> "Bicycle abandoned on the sidewalk at [PII_LOCATION]." — predicted as *veiligheid en omgeving* (*safety and environment*).

> "Improper placement of bicycles by a bicycle dealer at bicycle racks placed by the municipality." — predicted as *vervoer* (*transport*).

Vervoer (transport): This topic has 99 positive instances. The model made 70 errors, 24 FP and 46 FN. It is confused with *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*) (34 times), *veiligheid en omgeving* (*safety and environment*) (15 times) and *parkeren* (*parking*) (11 times). As with the other smaller topics, it is most often missed in favour of *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*). The examples are shown below.

> "The streetlight at the intersection is not working." — predicted as *veiligheid en omgeving* (*safety and environment*).

> "Car left on the sidewalk out of laziness. No passage for pedestrians possible anymore. We now have to go around via the bike path." — predicted as *parkeren* (*parking*).

Analysis of Subtopic Predictions

The per-topic results for the subtopics are reported in the tables of Section 5.3. The F1-scores cover a wide range, from 0.00 to 0.85. Subtopics that are frequent in the training set tend to reach a high recall, sometimes at the cost of precision. Subtopics that are rare in the training set tend to have a low recall, since the model rarely predicts them. The clearest examples are *geluidsoverlast* (*noise nuisance*), which reaches a recall of 0.94 against a precision of 0.27, and *criminaliteit* (*criminality*), which reaches a precision of 0.78 against a recall of 0.03. The confusions described here are therefore reported only for the subtopics of the main topics described above.

Subtopics of Opruimen, afval en onderhoud: This main topic contains the four most frequent subtopics: *onderhoud omgeving* (*environment maintenance*) (430 instances, 98 FP, 221 FN), *vuil/ongedierte overlast* (*waste/pest nuisance*) (355 instances, 154 FP, 145 FN), *bruikbaarheid/beschikbaarheid afvalcontainers* (*usability/availability of waste containers*) (163 instances, 83 FP, 91 FN) and *ophalen huiscontainers/grofvuil* (*collection of household containers/bulky waste*) (153 instances, 114 FP, 64 FN). The dominant pattern is that these four subtopics are confused with one another. *Onderhoud omgeving* is confused with *vuil/ongedierte overlast* 143 times, *vuil/ongedierte overlast* with *ophalen huiscontainers/grofvuil* 80 times and with *bruikbaarheid/beschikbaarheid afvalcontainers* 57 times, and *bruikbaarheid/beschikbaarheid afvalcontainers* with *ophalen huiscontainers/grofvuil* 58 times. The four subtopics describe closely related aspects of waste and maintenance, and the model often predicts one of them in place of another.

> "Next to the container in [PII_LOCATION] there have been full rubbish bags on the street for days." — the actual labels are *ophalen huiscontainers/grofvuil* (*collection of household containers/bulky waste*) and *vuil/ongedierte overlast* (*waste/pest nuisance*), predicted only *vuil/ongedierte overlast* (*waste/pest nuisance*).

Subtopics of Veiligheid en omgeving: The most frequent subtopic of this main topic is *criminaliteit* (*criminality*) (280 instances, 2 FP, 273 FN), which the model almost never predicts. When it should be assigned, the model predicts *geluidsoverlast* (*noise nuisance*) (101 times), *vuil/ongedierte overlast* (*waste/pest nuisance*) (99 times) or *algemene veiligheid* (*general safety*) (63 times). The criminal act is usually not stated directly but has to be inferred from an observable element, such as waste or damage, and the model predicts that observable element instead. The other subtopics show two opposite behaviours. *Algemene veiligheid* (112 instances, 108 FP, 59 FN) and *geluidsoverlast* (36 instances, 92 FP, 2 FN) are over-predicted: *geluidsoverlast* in particular is assigned more often than it appears. It is assigned in place of *criminaliteit*, *overlast personen* (*nuisance from people*) (47 times) and *geuroverlast* (*odour nuisance*) (28 times). *Geuroverlast* (49 instances, 3 FP, 48 FN) shows the opposite behavior, being missed almost every time, in favour of *geluidsoverlast*. *Verlichting* (*lighting*) (82 instances, 15 FP, 10 FN) is the most reliable subtopic of this group, with a low error count.

> "The fireworks completely destroyed the bin at the playground." —Actual labels *criminaliteit* and *vuil/ongedierte overlast*, predicted only *vuil/ongedierte overlast*.

Subtopics of Vervoer: This main topic contains *infrastructuur* (*infrastructure*) (72 instances, 28 FP, 33 FN), *verkeersmaatregelen* (*traffic measures*) (40 instances, 6 FP, 37 FN) and *verkeersovertreding* (*traffic violation*) (8 instances, 9 FP, 7 FN). The support of *verkeersovertreding* is too low for its confusions to be interpretable. The main pattern is between the other two: *verkeersmaatregelen* is confused with *infrastructuur* 24 times. The two describe related aspects of the road, and a request such as a speed bump can be read either as a traffic measure or as an infrastructure element, so the model predicts one in place of the other.

> "Could a speed bump be installed in [PII_LOCATION], people drive much too fast." — actual label *verkeersmaatregelen*, predicted *infrastructuur*.

Subtopics of Parkeren: The two most frequent subtopics are *parkeeroverlast* (*parking nuisance*) (79 instances, 32 FP, 21 FN) and *bereikbaarheid/beschikbaarheid parkeerplek* (*Accessibility/availability of parking spaces*) (49 instances, 31 FP, 37 FN). *Bereikbaarheid/beschikbaarheid parkeerplek* is confused with *parkeeroverlast* 29 times, which is a confusion between two close subtopics. *Parkeeroverlast* is the more frequent of the two and is the one the model tends to predict.

> "There is no free spot to be found in [PII_LOCATION], parking is a problem." — actual label *bereikbaarheid/beschikbaarheid parkeerplek*, predicted *parkeeroverlast*.

5.4.2 Quantitative Analysis for Re-Annotated test set

BERTje$_{\text{LLM}}$ is now analysed on the re-annotated test set. As discussed in Section 3.5, the full human-validated test set is biased towards the rule-based labels, so this set should be less affected by it. The goal here is to verify whether the confusions observed on the full test set still hold, and whether the balance between false negatives and false positives changes. The re-annotated test set contains 214 main topic labels and 213

subtopic labels in total, with a support per topic between 1 and a few dozen instances. The errors are reported both as totals (Table 5.9) and, for the most frequent main topics, topic by topic (Table 5.10).

Table 5.9: False negatives (FN), false positives (FP), and the share of false positives over total errors, on the full human-validated test set and on the re-annotated test set.

Test set	Level	FN	FP	FP share
Full	Main	366	386	51%
Full	Sub	1249	988	44%
Re-annotated	Main	69	96	58%
Re-annotated	Sub	132	132	50%

Two observations follow from the comparison.

- On the full test set the model tends to under-predict, especially at the subtopic level, where the false negatives outnumber the false positives (1,249 against 988, an FP share of 44%). At the main topic level the two are closer to balance (51%).
- On the re-annotated test set the balance moves towards over-prediction. At the main topic level the false positives outnumber the false negatives (96 against 69, an FP share of 58%), and at the subtopic level the two become balanced (132 each, 50%). At both levels the FP share is higher than on the full test set.

The direction of the errors therefore changes between the two test sets. On the reports the rule-based system classifies correctly, which make up most of the full test set, the model tends to assign fewer labels than the test set indicates. On the reports the rule-based system gets wrong, which are the harder and less typical cases, the model tends to assign more labels than the test set indicates. The error patterns observed on the full test set are not simply an artefact of the bias, since the model keeps confusing the same groups of topics (as the example below illustrates), but the balance between missed and incorrectly predicted labels shifts towards over-prediction on the re-annotated test set.

Analysis of Main Topic predictions of the Re-annotated test set

The main topic errors are reported in Table 5.10, next to the corresponding values on the full test set.

Table 5.10: False negatives (FN) and false positives (FP) per main topic on the full and on the re-annotated test set, for the most frequent main topics.

	Full		Re-annotated	
Main topic	**FN**	**FP**	**FN**	**FP**
opruimen, afval en onderhoud	39	148	6	36
dagelijks leven en sociale gelegenheden	96	64	43	5
veiligheid en omgeving	147	109	4	43
parkeren	26	31	6	6
vervoer	46	24	4	5

Two observations follow.

- The topics involved in the errors are the same on both test sets. *Opruimen, afval en onderhoud* (*cleaning, waste and maintenance*), *veiligheid en omgeving* (*safety and environment*) and *dagelijks leven en sociale gelegenheden* (*daily life and social occasions*) are the topics most often confused on the full test set (see Table 5.8), and they remain so on the re-annotated test set. The confusions described for the full human-validated test reappear on the re-annotated test set:

  > "Trash can destroyed" — the actual label is *veiligheid en omgeving* (*safety and environment*), but the model predicted *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*).

- Errors change for *veiligheid en omgeving* (*safety and environment*). On the full test set it is mostly missed (147 FN against 109 FP), while on the re-annotated test set it is mostly over-predicted (43 FP against 4 FN). On the re-annotated test set, BERTje$_{\text{LLM}}$ assigns this topic far more often than the test set indicates, the opposite of its behaviour on the full test set. On the other hand, *dagelijks leven en sociale gelegenheden* (*daily life and social occasions*) has 51 positive instances. The model made 48 errors, 5 FP and 43 FN. It is most often confused with *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*) (24 times) and *veiligheid en omgeving* (*safety and environment*) (27 times), following the prediction pattern on the full human-validated test set. With 43 FN over 51 instances, the topic is missed almost every time, confirming the pattern seen on the full human-validated test set (examples below).

  > "Every day fireworks are set off in the neighborhood. Very annoying since I haven't been able to walk my dogs since October!.") — the actual labels are *veiligheid en omgeving* (*safety and environment*) and *dagelijks leven en sociale gelegenheden* (*daily life and social occasions*), but the model predicted only *veiligheid en omgeving* (*safety and environment*).

A likely cause of this change is the composition of the re-annotated test set. This is discussed in the qualitative analysis (Section 5.4.3).

Analysis of Subtopic predictions of the Re-annotated test set

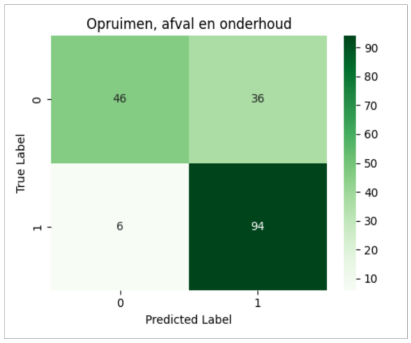
At the subtopic level the same shift towards over-prediction is visible. Two subtopics are missed recurrently:

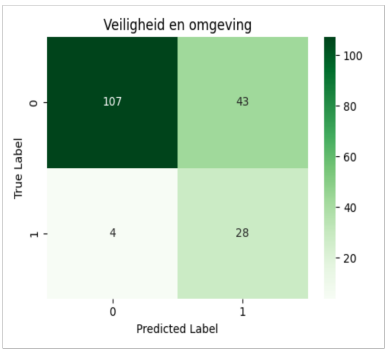
- *schade en claims* (*damage and claims*) (41 FN, 2 FP), confused mostly with *algemene veiligheid* (*general safety*) (9 times) and *vuil/ongedierte overlast* (9 times).

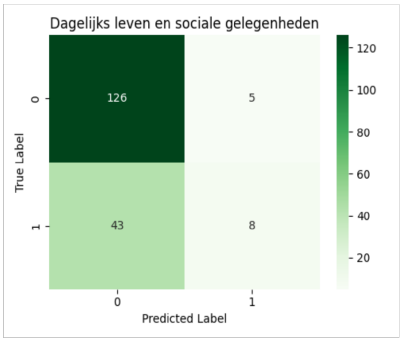
- *Onderhoud omgeving* (31 FN, 19 FP), confused mostly with *vuil/ongedierte overlast* (17 times) and *bruikbaarheid/beschikbaarheid afvalcontainers* (8 times).

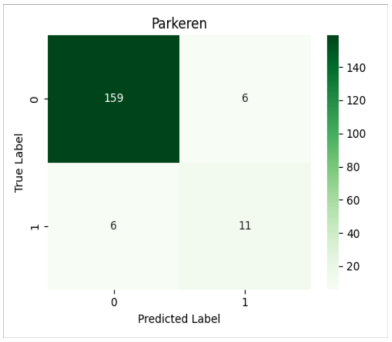
Examples can be found below.

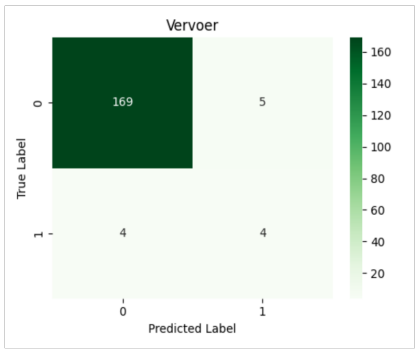
> "Bin blown up by fireworks" — the actual label is *schade en claims* (*damage and claims*), but the model predicted *algemene veiligheid* (*general safety*).











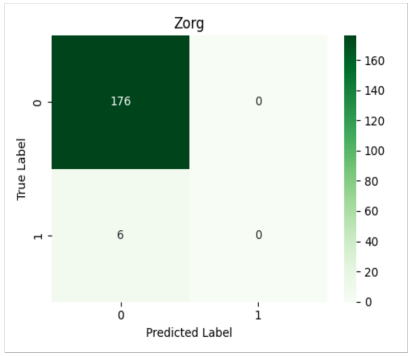


Figure 5.2: Confusion matrices of main topics for BERTje$_{\text{LLM}}$ (re-annotated test set). *Zorg* is added for completeness but is not present in the re-annotated test set error analysis.

> "[PII_LOCATION] trash can, lots of fireworks remains" — the actual label is *onderhoud omgeving* (*environment maintenance*), but the model predicted *vuil/ongedierte overlast* (*waste/pest nuisance*).

Some specific subtopics are over-predicted, in particular *vuil/ongedierte overlast* (45 FP, 1 FN), *bruikbaarheid/beschikbaarheid afvalcontainers* (28 FP, 0 FN) and *algemene veiligheid* (20 FP, 2 FN). As on the full test set, the model over-assigns the frequent and concrete subtopics and misses the ones that have to be inferred from the report.

5.4.3 Qualitative Analysis

The quantitative analysis focused on the preferred model, BERTje$_{\text{LLM}}$. It described the confusion patterns for main topics and subtopics on the full human-validated test set, and then on the re-annotated test set, where the same patterns were found. This qualitative analysis groups these patterns, discusses their possible causes, and suggests how they could be addressed. To identify these patterns, a sample of misclassified reports from the full human-validated test set was inspected manually. The inspection was exploratory: it aimed to identify recurrent error types rather than to quantify their frequency.

Overlapping Topics

A large part of the errors does not reflect a failure of the model, but an overlap between topics that are close in content. The clearest case is the overlap between *vervoer* (*transport*) and the other topics. The official description of *vervoer* covers road conditions and infrastructure issues, such as damaged crosswalks or road signs, and it states that part of its content may overlap with *veiligheid en omgeving* (*safety and environment*) (see the topic description in Section 3.2). The label *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*) adds a second overlap, since it contains maintenance (*onderhoud*), which is close to the road-condition content of *vervoer*. A report about the physical state of a road can therefore be read as *vervoer*, as *opruimen, afval en onderhoud*, or as *veiligheid en omgeving*, and the model predicts one while the test set indicates another. Two examples of this overlap are found below.

> "The bike path tiles are sinking and causing dangerous ruts." — predicted as *vervoer* (*transport*).

> "Street name sign at [PII_LOCATION] is twisted." — predicted as *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*).

The same overlap is present at the subtopic level, inside *opruimen, afval en onderhoud*, where *onderhoud omgeving* (*environment maintenance*), *vuil/ongedierte overlast* (*waste/pest nuisance*) and *ophalen huiscontainers/grofvuil* (*collection of household containers/bulky waste*) might apply to closely related aspects of waste and maintenance and are exchanged with one another, as shown in the quantitative analysis. A possible way to address this is to merge the topics and subtopics that overlap most, or to redefine their boundaries, so that the distinction is clearer for both the model and the annotator. However, the municipalities do not plan to merge topics, as they prefer to keep a fine-grained taxonomy.

Prediction of the observable element instead of the underlying issue

A second pattern is that the model tends to predict the concrete element described in the report rather than the issue behind it. When a report mentions a specific object or event, the model assigns the topic of that object, even when the actual problem is a different one. As shown in the example below.

> "Youths are constantly vandalizing bicycles in the underground parking garage." — predicted as *parkeren* (*parking*).

In the report the observable element is the "parking garage", so the model predicts *parkeren*, while the issue is the vandalism by youth. This is the same under-prediction of *criminaliteit* described in the quantitative analysis. For instance, several reports talk about trash bins being blown up by fireworks. Here, the model learns the pattern that connects trash bin to cleaning, often assigning *vuil/ongedierte overlast* (*waste/pest nuisance*) as subtopic and discarding the criminal act. The model therefore captures what is named in the text but not what has to be inferred from it. Addressing this would require training data where the label reflects the issue rather than the observable element, which in turn depends on the quality of the annotations, an issue addressed in the next subsection.

Quality of the Annotations

A third important problem concerns the test set itself. The labels of the full human-validated test set were assigned by the rule-based system and validated by marking them as correct or incorrect, and the rule-based system assigns topics on the basis of keywords. As a result, some labels in the test set are questionable, and several report labels counted as errors are in fact cases where the model is at least as plausible as the test set label.

> "Trash can broken." — actual label *dagelijks leven en sociale gelegenheden* (*daily life and social occasions*), predicted *opruimen, afval en onderhoud* (*cleaning, waste and maintenance*).

> "Dog poop bin blown up." — actual label *dagelijks leven en sociale gelegenheden* (*daily life and social occasions*), predicted *veiligheid en omgeving* (*safety and environment*).

> "Trash cans near the soccer field were blown up. Broken bottles on the path leading down." — actual labels *Veiligheid en omgeving* (*Safety and environment*) with subtopic of *criminaliteit*, *Dagelijks leven en sociale gelegenheden* with subtopic *buurt ambiance* (*Neighborhood ambiance*) and *Opruimen, afval en onderhoud* (*Cleaning, waste and maintenance*) with subtopic *Vuil/ongedierte overlast* (*Waste/pest nuisance*).

In the first two cases the actual set label is hard to justify, and the prediction of the model is at least as reasonable. A keyword such as the name of a place or object can lead the rule-based system to assign a topic that does not match the content of the report, and these labels are then kept in the test set. It is also visible in the third example: safety concerns and cleaning might be reasonable for that report, however, it is harder to justify the neighborhood ambiance. This pattern was noticed during the manual

inspection of the reports, but the frequency of test set labels that can be considered incorrect was not quantified. The same problem with a rule-based system in the same domain was already reported by Szabo (2024), who inspected a sample of the test set and found that a notable percentage of the labels assigned by the rule-based system were considered erroneous (around 38% of the sample). Using a test set validated on such a system is therefore not recommended. A more reliable evaluation would require a test set annotated by a group of human experts (or trained on the matter) following clear annotation guidelines, with inter-annotator agreement to measure the consistency of the labels.

Binary Relevance and Label Co-occurrence

A last issue comes from the problem transformation used for the classification. The *binary relevance* transformation treats each topic as a separate binary problem, so the model cannot learn that certain topics tend to appear together. Topics that co-occur frequently, such as the subtopics of *opruimen, afval en onderhoud*, are predicted independently, and the model misses the cases where one of them implies the presence of another. A problem transformation that considers label co-occurrence could improve the prediction of the reports that carry more related labels.

5.4.4 Summary

The error analysis shows that the errors of the model are concentrated in a small number of topics and follow the same patterns on both test sets. A large part of these errors does not reflect a failure of the model but an overlap between topics that are close in content, a tendency to predict the observable element instead of the issue behind it, and a test set whose labels are not always reliable. In particular, the issues with the human-validated test set and the re-annotated test set are crucial. Both test sets raise reliability and validity concerns: the absence of inter-annotator agreement to measure annotation consistency, a full test set composed in large part of reports aligned with the rule-based system (1,121 of 1,303), and the small size of the re-annotated test set (182 reports). Moreover, the binary relevance transformation adds a further limitation, since it prevents the model from learning which topics tend to appear together.

These observations point to two directions that are taken up in the next chapter: a revision of the taxonomy, merging or redefining the topics that overlap most, and a test set annotated by humans with clear guidelines, which may allow a more reliable evaluation than the one based on the labels of the rule-based system.

Chapter 6

Discussion & Conclusion

6.1 Concluding Remarks

This thesis studied multi-label topic classification of Dutch citizens' reports in the governance domain, in a context of scarcity of annotated data (Section 2.3). The aim was to compare two ways of addressing this problem, namely the zero-shot prediction of a generative language model and an encoder model fine-tuned on labels generated by the best performing generative language model. Three experiments were carried out: a BERTje model fine-tuned on the rule-based labels ($\text{BERTje}_{\text{RB}}$), used as a reference point; the zero-shot prediction of GPT-4.1 and Llama 3.3 70B Instruct, evaluated with a one-step and a two-step classification approach (Section 4.1.3); and a BERTje model fine-tuned on the labels generated by the best approach, GPT-4.1 in the one-step classification approach ($\text{BERTje}_{\text{LLM}}$). All models were evaluated on a full human-validated test set and on a re-annotated test set, a subset of the full one. The main finding is that $\text{BERTje}_{\text{LLM}}$ reaches a performance comparable to the zero-shot prediction of GPT-4.1 and is therefore recommended for operational use within the company.

However, the composition of the full human-validated test set is biased towards the rule-based labels. As described in Section 3.5, the domain expert only validated the labels already assigned by the rule-based system, marking them as correct or incorrect, and most of the test set consists of reports that the system classifies correctly. The re-annotated test set was built to address this issue. The reports marked as incorrect were re-annotated by hand, providing a re-annotated test set whose labels are less dependent on the rule-based system.

On the full human-validated test set, $\text{BERTje}_{\text{RB}}$ obtains the highest scores, with a micro-F1 of 0.86 on the main topics and 0.77 on the subtopics. Nevertheless, this does not reflect an actual outperformance of the model: the re-annotated test set confirms this. The same model drops to a micro-F1 of 0.57 on the main topics and 0.34 on the subtopics, a much larger drop than the other approaches, which fall by 0.12–0.14 on the main topics and around 0.08 on the subtopics. When evaluated with the re-annotated test set, the model that learned patterns of the rule-based system reproduces them.

In this study, one of the main comparisons is the one between the zero-shot prediction of GPT-4.1 with one-step classification approach (which also generates the labels) and $\text{BERTje}_{\text{LLM}}$. On the full test set the two are comparable on the main topics (0.77 micro-F1 for GPT-4.1, 0.78 for $\text{BERTje}_{\text{LLM}}$) and on the subtopics (0.48 against 0.46). On the re-annotated test set they almost coincide: 0.65 and 0.64 on the main topics, and 0.38 for both on the subtopics. The encoder reproduces the predictive behaviour of

GPT-4.1 on the test set and on the re-annotated test set. This shows that an encoder model can be trained on LLM-predicted labels, which offers a viable approach to the scarcity of annotated data.

Given this comparable performance, the choice between the two approaches depends on operational considerations. BERTje can be run locally without depending on an external service, it has a much lower computational cost, and it is faster. Zero-shot prediction with a generative language model instead requires access through an API: this involves a cost for every request, it depends on an external service, and it is subject to content safety filters that can discard some reports. Once the LLM-predicted labels have been generated and the encoder has been trained, the generative model is no longer needed. For an operational deployment in the company, $\text{BERTje}_{\text{LLM}}$ may be the preferable option.

This result must be read together with the error analysis. Multi-label topic classification is difficult, and the subtopic level is the harder part of the taxonomy, being more fine-grained and containing many rare classes, with micro-F1 scores between 0.38 and 0.48. The error analysis showed, however, that a part of these errors does not reflect a failure of the model but an overlap between topics that are close in content and a number of test set labels that are themselves not always justifiable (Section 5.4.3). The subtopic level would benefit from a clearer taxonomy and a more reliable test set, as discussed in Section 5.4.3.

6.1.1 Research Questions

Main Question: Which approach yields the best performance for multi-label topic classification in the governance domain: zero-shot prediction using generative language models or an encoder model fine-tuned on LLM-predicted labels?

Both approaches reach comparable performance. On the full human-validated test set, GPT-4.1 and $\text{BERTje}_{\text{LLM}}$ are almost equivalent on the main topic classification (0.77 against 0.78 micro-F1) and on the subtopic classification (0.48 against 0.46). On the re-annotated test set their results also almost coincide (0.65 against 0.64 on the main topics, 0.38 for both on the subtopics). In light of this, the answer cannot be based on the scores alone, but on the characteristics of the two approaches: $\text{BERTje}_{\text{LLM}}$ is preferable for the operational reasons given in the previous section. $\text{BERTje}_{\text{RB}}$, although it obtained the highest results on the full human-validated test set, does not represent a valid comparison, because of the selection bias of the test set (Section 3.5).

Subquestion 1: What is the predictive performance of zero-shot generative language models for multi-label topic classification on Dutch governance data when evaluated on a human-validated test set?

GPT-4.1 reaches a micro-F1 of 0.77 on the main topics on the full human-validated test set and 0.65 on the re-annotated test set, and 0.48 and 0.38 on the subtopics for both sets respectively. It outperforms Llama 3.3 70B Instruct in the one-step classification, where Llama reaches 0.51 on the main topics and 0.19 on the subtopics on the full human-validated test set, and 0.60 on the main topics and 0.33 on the subtopics on the re-annotated test set. The comparison between one-step and two-step does not point to a single best approach across both models: GPT-4.1 is best or comparable in the one-step classification in most cases, whereas Llama improves substantially in

the two-step classification (from 0.51 to 0.73 on the main topics and from 0.19 to 0.48 on the subtopics for the full human-validated test set). GPT-4.1 in the one-step classification approach was selected to generate the LLM-predicted labels. The one-step classification approach is also a cheaper solution for generating labels.

Subquestion 2: What is the predictive performance of a transformer-based encoder fine-tuned on LLM-predicted labels for multi-label topic classification on Dutch governance data when evaluated on the same human-validated test set?

$\text{BERTje}_{\text{LLM}}$ reaches a micro-F1 of 0.78 on the main topics and 0.46 on the subtopics on the full human-validated test set, and 0.64 and 0.38 on the re-annotated test set. The error analysis showed that part of these errors, especially at the subtopic level, does not reflect a failure of the model but the overlap between topics that are close in content and test set labels that are themselves not always justifiable (Section 5.4.3). These scores must be read together with the bias of the full human-validated test set discussed above: they should not be taken as an absolute measure of the model's quality, but as evidence that the encoder reaches the level of the model that produced its labels, while being smaller and cheaper.

6.2 Limitations

One of the main limitations of this work concerns, as stressed in multiple occasions, the composition of the human-validated test set. The results on the full human-validated test set must be read with care, and $\text{BERTje}_{\text{RB}}$ cannot be considered a fair competitor (Section 6.1). The re-annotated test set, consisting of reports the rule-based system misclassified, was introduced precisely to provide an evaluation portion as independent as possible from this bias, but it represents only a small part of the whole test set, covering 6 of the 10 main topics and 18 of the 37 subtopics, so the two test sets are not fully comparable. With only 182 reports, its scores are also sensitive to individual reports.

On the training side, the training set remains highly imbalanced even after the undersampling of the dominant class. The undersampling reduced the frequency of the most represented topic but did not remove the overall imbalance, which continues to penalize the rare topics. The LLM-predicted labels carry a further limitation. GPT-4.1 tends to avoid some topics, and this tendency is reflected in the LLM-predicted label distribution (see Tables 3.3 and 3.4). Labels and passed on to the BERTje model fine-tuned on them, so the quality of the LLM-predicted labels is bound to the biases of the generative language model used. Finally, because of time and computational constraints, no hyperparameter tuning was performed, and BERTje follows the standard configuration recommended for fine-tuning in (Devlin et al., 2018).

6.3 Future Work

The directions for future work follow from the limitations just discussed. The first concerns the creation of a test set without the selection bias present in this work. Rather than a balanced test set, a set that represents the whole taxonomy and is independent of the rule-based system, would already allow a fairer comparison. Annotating the test set with several annotators and measuring inter-annotator agreement would also give

insight into the cases that are genuinely difficult to label (as discussed in Subsection 5.4.3).

As for the training data, the imbalance could be addressed with oversampling or data augmentation techniques, and combining these with LLM-predicted labels would be a natural extension of this work. Working with spontaneous reports offers an advantage here: citizens' reports keep arriving through the official governmental channels and are available in large quantities, so with a more careful selection and preprocessing of additional reports it would be possible to build a dataset with more balanced classes.

As for the use of the generative language models, both for the zero-shot prediction and for the generation of the labels used in training, more structured and detailed prompts could make the model more precise. Using few-shot prompting instead of zero-shot for the generation of the LLM-predicted labels would make it possible to assess whether a small number of examples in the prompt improves the quality of the labels and the performance of the encoder trained on them.

A last direction concerns the taxonomy. The error analysis revealed a systematic overlap between topics that are close in content, both at the main topic and at the subtopic level. A revision of the taxonomy could reconsider whether some of these overlapping topics should be merged or more clearly delimited, which would trade part of the granularity of the labels for more reliable predictions.

More broadly, the findings suggests that for specialised domains where manual annotation is scarce and costly, LLM-predicted labels might be a practical way to train a smaller, encoder-based classifier that does not heavily depend on a commercial API.

Appendix A

Appendix

Data Statement

This appendix provides a data statement for the dataset used in this thesis, following the framework introduced by Bender and Friedman (2018). The goal of a data statement is to document a dataset so that developers and users have the information needed to judge to what extent results obtained on it may generalize, and to make explicit the biases and ethical considerations attached to it. The data statement template was taken from "Data Statement for NLP"[1].

Data set name: "MarketResponse citizens' reports (internal dataset, not publicly released)".

Citation: N/A. The dataset is not publicly released and is not associated with a citation.

Data set developer: The dataset was assembled and anonymized by the partner company MarketResponse, which receives and processes the reports on behalf of Dutch municipalities.

Data statement author: Gianmarco Sapienza.

Curation rationale

The dataset consists of reports submitted by Dutch citizens to municipalities, collected and anonymized by the partner company MarketResponse. The reports are produced through the municipalities' regular channels, mainly three:

- website forms
- email
- transcribed phone calls

[1] https://sites.google.com/uw.edu/data-statements-for-nlp/sample-data-statements

No selection was made on the content of the reports beyond the removal of duplicates and the undersampling of the dominant topic described in Chapter 3. The label space is organized into a taxonomy of 10 main topics and 37 subtopics (Please refer to Figure 3.1).

Language variety

- **BCP-47 tag:** `nl`.
- **Description:** Dutch as written by citizens of Dutch municipalities, in spontaneous and informal communication addressed to the local authority. The original Dutch text is preserved throughout the experiments and the model used for fine-tuning relies entirely on Dutch-native resources. The reports are complaints and concerns raised by citizens. The language informal and linguistically diverse, and includes spelling and grammatical errors typical of user-generated text.

Speaker demographic

The reports are written by citizens addressing their municipality. No demographic information about the authors is recorded by the partner company and none is used in this work.

- **Age:** unknown.
- **Gender:** unknown.
- **Race/ethnicity:** unknown.
- **First language(s):** unknown; presumed to be predominantly Dutch.
- **Socioeconomic status:** unknown.
- **Number of different speakers represented:** unknown.
- **Presence of disordered speech:** unknown.

Although no demographic information is available at the individual level, the reports are not mixed across municipalities: each corpus comes from a specific municipality covering specific neighbourhoods. The demographic composition of those neighbourhoods may therefore act as a latent variable in the text, even though it is not recorded.

Annotator demographic

The dataset is not annotated by human annotators in the traditional sense. The training labels are produced automatically by two different methods: the partner company's rule-based system and the generative language model GPT-4.1. Test set was human-validated a domain expert at the partner company, who performed a binary validation of the rule-based predictions (Chapter 3.5). Another test set is used in this work to overcome biases of the human-validation: The re-annotated test set was annotated by the author of this thesis.

- **Domain expert at the partner company:** demographic information not disclosed. Training: domain expert in the classification of Dutch citizens' reports for the partner company.

- **Author of this thesis (annotators of the re-annotated test set):**
  - age: 28
  - first language: Italian
  - Training: MA Student in Linguistics (Language & AI)

  Not a native speaker of Dutch; the re-annotation was carried out on English translations of the reports, as discussed in Section 3.5.1.

Speech situation

- **Time and place:** the reports were collected by Dutch municipalities through the partner company. The exact time window is not disclosed by the company.
- **Modality:** written.
- **Scripted/edited vs. spontaneous:** spontaneous. The reports are written by citizens on their own initiative.
- **Synchronous vs. asynchronous interaction:** asynchronous. The citizen writes the report and the municipality processes it afterwards.
- **Intended audience:** the municipality, that is, the local authority responsible for handling the issue raised.

Text characteristics

The reports address topics related to urban life and to the problems that arise in it: cleanliness and waste collection, parking, mobility, safety, neighborhood atmosphere, administrative requests, and so on. The vocabulary is governance-specific, it serves as a means of bringing the issues to the attention of municipal authorities. The length of the reports varies considerably, from very short messages of fewer than 50 characters to multi-sentence descriptions of more articulated problems.

Recording quality

N/A. The dataset consists only of written text.

Other: anonymization, privacy, and ethical considerations

Before the data are handed over to the author of this thesis, they pass through an anonymization system run by the partner company, which replaces Personally Identifiable Information (PII) with placeholder tokens (for example `[PII_LOCATION]`). This reduces the risk of direct identification of individuals. However, anonymization does not remove every trace of the people writing the reports: style, location references, and the nature of the complaint can still carry indirect information. This thesis does not use any sensitive information or reports containing explicitly identifiable information. All experiments were carried out on equipment provided by the partner company, and the dataset is not distributed.

Three sources of bias should be made explicit:

- **Residual bias after anonymization.** The removal of PII reduces direct identification but does not remove latent cultural and gender biases that may be present in the linguistic patterns of the reports.
- **Geographic and demographic bias.** Because each corpus belongs to a specific municipality and set of neighbourhoods, biases linked to the demographics of those areas may be reflected in the language and, in turn, in the labels learned by the models.
- **Test set bias.** The test set is itself biased. It was built from the labels assigned by the rule-based system, and the large majority of its reports are cases that the system already classifies correctly. Any model that has learned the patterns of the rule-based system is therefore favoured on this test set, and the results obtained on it must be read with caution. The re-annotated test set was introduced to provide a portion of the evaluation that is less dependent on the rule-based system, but it covers only a part of the taxonomy.

A further consideration concerns the labels themselves: the training labels are produced automatically, either by the rule-based system or by a generative language model, and any systematic bias of these sources, such as the tendency of GPT-4.1 to underpredict sensitive topics, is passed on to the models trained on them. Because the data describe citizens' problems, a model that systematically ignores certain topics could, if deployed without supervision, lead some categories of reports to be overlooked.

Provenance appendix

N/A. The dataset is not built on top of an existing publicly released dataset.

A.1 Result Tables

In this section are found the extensive results for Llama 3.3 70B Instruct direct zero-shot prediction one-step and two-step, GPT-4.1 two-step, BERTje fine tuned with rule-based labels and BERTje fine-tuned with LLM-predicted labels.

A.1.1 GPT-4.1 (one-step)

Table A.1: GPT-4.1 (one-step) – main topics, full test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.79	0.67	0.73	515
opruimen, afval en onderhoud	0.84	0.94	0.89	799
dagelijks leven en sociale geleg.	0.37	0.21	0.27	136
parkeren	0.81	0.77	0.79	123
vervoer	0.64	0.43	0.52	99
zorg	0.83	0.42	0.56	12
financiële ondersteuning	0.00	0.00	0.00	2
wonen en ondernemen	0.50	0.50	0.50	2
burgerzaken	1.00	0.50	0.67	6
maatschappelijke ondersteuning	0.00	0.00	0.00	2
micro avg	0.79	0.75	0.77	1696
macro avg	0.58	0.45	0.49	1696

Table A.2: GPT-4.1 (one-step) – subtopics, full test set.

Class	P	R	F1	Supp.
criminaliteit	0.88	0.07	0.14	280
algemene veiligheid	0.30	0.42	0.35	112
verlichting	0.82	0.80	0.81	82
overlast personen	0.45	0.36	0.40	75
geluidsoverlast	0.32	0.92	0.48	36
geuroverlast	0.33	0.02	0.04	49
vuil/ongedierte overlast	0.60	0.64	0.62	355
onderhoud omgeving	0.64	0.45	0.52	431
bruikb./beschikb. afvalcontainers	0.56	0.60	0.58	163
huisdierenoverlast	0.20	0.83	0.32	6
buurt ambiance	0.18	0.05	0.08	58
evenementen, feestd. en herd.	0.05	1.00	0.09	1
hangjongeren	0.90	0.90	0.90	10
ophalen huiscont./grofvuil	0.46	0.53	0.49	153
parkeergarage	0.80	0.31	0.44	15
verkeersmaatregelen	0.79	0.28	0.41	40
infrastructuur	0.49	0.35	0.41	72
bereikb./beschikb. parkeerplek	0.44	0.51	0.47	49
parkeeroverlast	0.63	0.71	0.67	79
verkeersveiligheid	0.23	0.60	0.33	25

Class	P	R	F1	Supp.
verkeersovertreding	0.17	0.12	0.14	8
schade en claims	0.49	0.34	0.40	64
burgemeester	0.67	0.67	0.67	3
hulp aan dak- en thuislozen	0.83	0.83	0.83	6
bijstand	0.00	0.00	0.00	1
huur en verhuur	0.50	0.50	0.50	3
verhuizen	1.00	0.50	0.67	4
parkeervergunning/abonnement	1.00	0.29	0.44	7
parkeerkosten	0.00	0.00	0.00	2
documentaanvraag	0.00	0.00	0.00	1
hulpmiddelen en woonvoorz.	0.00	0.00	0.00	2
financiële regelingen	0.00	0.00	0.00	1
vaccinatie	0.00	0.00	0.00	1
luchtkwaliteit	0.50	1.00	0.67	1
parkeerterminal	0.00	0.00	0.00	1
parkeren bezoekers	1.00	1.00	1.00	2
overlijden	1.00	1.00	1.00	1
micro avg	0.52	0.45	0.48	2195
macro avg	0.47	0.45	0.40	2195

Table A.3: GPT-4.1 (one-step) – main topics, challenge test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.44	0.88	0.58	32
opruimen, afval en onderhoud	0.73	0.94	0.82	100
dagelijks leven en sociale geleg.	0.55	0.12	0.19	51
parkeren	0.72	0.76	0.74	17
vervoer	0.44	0.50	0.47	8
zorg	0.50	0.17	0.25	6
micro avg	0.63	0.68	0.65	214
macro avg	0.56	0.56	0.51	214

Table A.4: GPT-4.1 (one-step) – subtopics, challenge test set.

Class	P	R	F1	Supp.
criminaliteit	0.75	0.27	0.40	11
algemene veiligheid	0.22	0.55	0.32	11
overlast personen	0.36	0.67	0.47	6
geluidsoverlast	0.80	0.57	0.67	7
vuil/ongedierte overlast	0.09	0.83	0.16	6
onderhoud omgeving	0.50	0.39	0.44	54
buurt ambiance	0.33	0.33	0.33	3
evenementen, feestd. en herd.	0.50	1.00	0.67	1

Class	P	R	F1	Supp.
hangjongeren	0.00	0.00	0.00	1
ophalen huiscont./grofvuil	0.70	0.35	0.47	40
verkeersmaatregelen	0.50	0.50	0.50	2
infrastructuur	0.17	0.25	0.20	4
parkeeroverlast	0.85	0.69	0.76	16
verkeersveiligheid	0.17	1.00	0.29	2
verkeersovertreding	1.00	0.50	0.67	2
schade en claims	0.62	0.11	0.19	45
hulp aan dak- en thuislozen	0.50	1.00	0.67	1
parkeren bezoekers	1.00	1.00	1.00	1
micro avg	0.38	0.38	0.38	213
macro avg	0.50	0.56	0.45	213

A.1.2 Llama 3.3 70B Instruct (one-step)

Table A.5: Llama 3.3 70B Instruct (one-step) – main topics, full test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.42	0.57	0.48	515
opruimen, afval en onderhoud	0.65	0.70	0.67	799
dagelijks leven en sociale geleg.	0.19	0.09	0.12	136
parkeren	0.16	0.15	0.15	123
vervoer	0.17	0.11	0.13	99
zorg	0.00	0.00	0.00	12
financiële ondersteuning	0.00	0.00	0.00	2
wonen en ondernemen	0.00	0.00	0.00	2
burgerzaken	0.00	0.00	0.00	6
maatschappelijke ondersteuning	0.00	0.00	0.00	2
micro avg	0.49	0.53	0.51	1696
macro avg	0.16	0.16	0.16	1696

Table A.6: Llama 3.3 70B Instruct (one-step) – subtopics, full test set.

Class	P	R	F1	Supp.
criminaliteit	0.29	0.03	0.05	280
algemene veiligheid	0.13	0.21	0.16	112
verlichting	0.04	0.05	0.04	82
overlast personen	0.06	0.21	0.09	75
geluidsoverlast	0.09	0.53	0.15	36
geuroverlast	0.01	0.02	0.01	49
vuil/ongedierte overlast	0.29	0.46	0.36	355
onderhoud omgeving	0.34	0.33	0.34	431
bruikb./beschikb. afvalcontainers	0.12	0.21	0.15	163

Class	P	R	F1	Supp.
huisdierenoverlast	0.00	0.00	0.00	6
buurt ambiance	0.14	0.09	0.11	58
evenementen, feestd. en herd.	0.00	0.00	0.00	1
hangjongeren	0.00	0.00	0.00	10
ophalen huiscont./grofvuil	0.20	0.18	0.19	153
parkeergarage	0.00	0.00	0.00	15
verkeersmaatregelen	0.20	0.07	0.11	40
infrastructuur	0.12	0.08	0.10	72
bereikb./beschikb. parkeerplek	0.03	0.02	0.02	49
parkeeroverlast	0.14	0.20	0.16	79
verkeersveiligheid	0.04	0.24	0.07	25
verkeersovertreding	0.00	0.00	0.00	8
schade en claims	0.09	0.03	0.05	64
burgemeester	0.00	0.00	0.00	3
hulp aan dak- en thuislozen	0.00	0.00	0.00	6
bijstand	0.00	0.00	0.00	1
huur en verhuur	0.00	0.00	0.00	3
verhuizen	0.00	0.00	0.00	4
parkeervergunning/abonnement	0.00	0.00	0.00	7
parkeerkosten	0.00	0.00	0.00	2
documentaanvraag	0.00	0.00	0.00	1
hulpmiddelen en woonvoorz.	0.00	0.00	0.00	2
financiële regelingen	0.00	0.00	0.00	1
vaccinatie	0.00	0.00	0.00	1
luchtkwaliteit	0.00	0.00	0.00	1
parkeerterminal	0.00	0.00	0.00	1
parkeren bezoekers	0.50	0.50	0.50	2
overlijden	0.00	0.00	0.00	1
micro avg	0.17	0.22	0.19	2195
macro avg	0.08	0.09	0.07	2195

Table A.7: Llama 3.3 70B Instruct (one-step) – main topics, challenge test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.32	0.94	0.48	32
opruimen, afval en onderhoud	0.70	0.88	0.78	100
dagelijks leven en sociale geleg.	0.55	0.12	0.19	51
parkeren	0.71	0.59	0.65	17
vervoer	0.71	0.62	0.67	8
zorg	0.00	0.00	0.00	6
micro avg	0.55	0.65	0.60	214
macro avg	0.50	0.52	0.46	214

Table A.8: Llama 3.3 70B Instruct (one-step) – subtopics, challenge test set.

Class	P	R	F1	Supp.
criminaliteit	1.00	0.18	0.31	11
algemene veiligheid	0.27	0.82	0.41	11
overlast personen	0.14	0.83	0.24	6
geluidsoverlast	0.25	0.86	0.39	7
vuil/ongedierte overlast	0.06	0.83	0.12	6
onderhoud omgeving	0.42	0.50	0.46	54
buurt ambiance	0.14	0.33	0.20	3
evenementen, feestd. en herd.	0.00	0.00	0.00	1
hangjongeren	0.00	0.00	0.00	1
ophalen huiscont./grofvuil	0.67	0.35	0.46	40
verkeersmaatregelen	0.33	0.50	0.40	2
infrastructuur	0.40	0.50	0.44	4
parkeeroverlast	0.62	0.62	0.62	16
verkeersveiligheid	0.09	1.00	0.17	2
verkeersovertreding	0.00	0.00	0.00	2
schade en claims	0.40	0.04	0.08	45
hulp aan dak- en thuislozen	0.00	0.00	0.00	1
parkeren bezoekers	1.00	1.00	1.00	1
micro avg	0.27	0.41	0.33	213
macro avg	0.32	0.47	0.29	213

A.1.3 GPT-4.1 (two-step)

Table A.9: GPT-4.1 (two-step) – main topics, full test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.78	0.60	0.68	515
opruimen, afval en onderhoud	0.87	0.89	0.88	799
dagelijks leven en sociale geleg.	0.40	0.01	0.03	136
parkeren	0.86	0.76	0.81	123
vervoer	0.67	0.10	0.18	99
zorg	0.00	0.00	0.00	12
financiële ondersteuning	0.00	0.00	0.00	2
wonen en ondernemen	0.20	1.00	0.33	2
burgerzaken	0.00	0.00	0.00	6
maatschappelijke ondersteuning	0.00	0.00	0.00	2
micro avg	0.84	0.66	0.74	1696
macro avg	0.38	0.34	0.29	1696

Table A.10: GPT-4.1 (two-step) – subtopics, full test set.

Class	**P**	**R**	**F1**	**Supp.**
criminaliteit	0.80	0.16	0.26	280
algemene veiligheid	0.28	0.54	0.37	112
verlichting	0.82	0.66	0.73	82
overlast personen	0.49	0.33	0.40	75
geluidsoverlast	0.32	0.94	0.48	36
geuroverlast	0.00	0.00	0.00	49
vuil/ongedierte overlast	0.70	0.41	0.52	355
onderhoud omgeving	0.65	0.50	0.57	431
bruikb./beschikb. afvalcontainers	0.52	0.72	0.60	163
huisdierenoverlast	0.17	0.33	0.22	6
buurt ambiance	0.25	0.02	0.03	58
evenementen, feestd. en herd.	0.00	0.00	0.00	1
hangjongeren	0.36	0.50	0.42	10
ophalen huiscont./grofvuil	0.49	0.51	0.50	153
parkeergarage	0.75	0.23	0.35	15
verkeersmaatregelen	0.86	0.15	0.26	40
infrastructuur	0.50	0.07	0.12	72
bereikb./beschikb. parkeerplek	0.49	0.51	0.50	49
parkeeroverlast	0.62	0.76	0.69	79
verkeersveiligheid	0.27	0.68	0.38	25
verkeersovertreding	0.25	0.12	0.17	8
schade en claims	0.00	0.00	0.00	64
burgemeester	0.00	0.00	0.00	3
hulp aan dak- en thuislozen	0.00	0.00	0.00	6
bijstand	0.00	0.00	0.00	1
huur en verhuur	0.67	1.00	0.80	3
verhuizen	0.00	0.00	0.00	4
parkeervergunning/abonnement	1.00	0.29	0.44	7
parkeerkosten	0.00	0.00	0.00	2
documentaanvraag	0.00	0.00	0.00	1
hulpmiddelen en woonvoorz.	0.00	0.00	0.00	2
financiële regelingen	0.00	0.00	0.00	1
vaccinatie	0.00	0.00	0.00	1
luchtkwaliteit	0.00	0.00	0.00	1
parkeerterminal	0.00	0.00	0.00	1
parkeren bezoekers	0.67	1.00	0.80	2
overlijden	0.00	0.00	0.00	1
micro avg	0.53	0.41	0.46	2195
macro avg	0.32	0.28	0.26	2195

Table A.11: GPT-4.1 (two-step) – main topics, challenge test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.42	0.69	0.52	32
opruimen, afval en onderhoud	0.78	0.89	0.83	100
dagelijks leven en sociale geleg.	0.50	0.02	0.04	51
parkeren	0.87	0.76	0.81	17
vervoer	0.60	0.38	0.46	8
zorg	0.00	0.00	0.00	6
micro avg	0.68	0.60	0.64	214
macro avg	0.53	0.46	0.44	214

Table A.12: GPT-4.1 (two-step) – subtopics, challenge test set.

Class	P	R	F1	Supp.
criminaliteit	0.50	0.36	0.42	11
algemene veiligheid	0.18	0.55	0.27	11
overlast personen	0.33	0.50	0.40	6
geluidsoverlast	0.83	0.71	0.77	7
vuil/ongedierte overlast	0.15	0.67	0.24	6
onderhoud omgeving	0.65	0.56	0.60	54
buurt ambiance	0.00	0.00	0.00	3
evenementen, feestd. en herd.	0.00	0.00	0.00	1
hangjongeren	0.33	1.00	0.50	1
ophalen huiscont./grofvuil	0.72	0.33	0.45	40
verkeersmaatregelen	0.50	0.50	0.50	2
infrastructuur	0.00	0.00	0.00	4
parkeeroverlast	0.86	0.75	0.80	16
verkeersveiligheid	0.12	0.50	0.20	2
verkeersovertreding	1.00	0.50	0.67	2
schade en claims	0.00	0.00	0.00	45
hulp aan dak- en thuislozen	0.00	0.00	0.00	1
parkeren bezoekers	1.00	1.00	1.00	1
micro avg	0.45	0.38	0.42	213
macro avg	0.40	0.44	0.38	213

A.1.4 Llama 3.3 70B Instruct (two-step)

Table A.13: Llama 3.3 70B Instruct (two-step) – main topics, full test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.63	0.71	0.67	515
opruimen, afval en onderhoud	0.78	0.95	0.86	799
dagelijks leven en sociale geleg.	0.00	0.00	0.00	136
parkeren	0.80	0.73	0.76	123
vervoer	0.63	0.42	0.51	99
zorg	0.00	0.00	0.00	12
financiële ondersteuning	0.00	0.00	0.00	2
wonen en ondernemen	0.10	1.00	0.18	2
burgerzaken	0.00	0.00	0.00	6
maatschappelijke ondersteuning	0.00	0.00	0.00	2
micro avg	0.72	0.74	0.73	1696
macro avg	0.29	0.38	0.30	1696

Table A.14: Llama 3.3 70B Instruct (two-step) – subtopics, full test set.

Class	P	R	F1	Supp.
criminaliteit	0.80	0.19	0.30	280
algemene veiligheid	0.21	0.54	0.30	112
verlichting	0.79	0.55	0.65	82
overlast personen	0.32	0.49	0.39	75
geluidsoverlast	0.16	0.94	0.28	36
geuroverlast	0.05	0.04	0.05	49
vuil/ongedierte overlast	0.62	0.57	0.59	355
onderhoud omgeving	0.50	0.75	0.60	431
bruikb./beschikb. afvalcontainers	0.48	0.87	0.62	163
huisdierenoverlast	0.18	0.50	0.26	6
buurt ambiance	0.00	0.00	0.00	58
evenementen, feestd. en herd.	0.00	0.00	0.00	1
hangjongeren	0.62	0.50	0.56	10
ophalen huiscont./grofvuil	0.49	0.47	0.48	153
parkeergarage	1.00	0.31	0.47	15
verkeersmaatregelen	0.87	0.33	0.47	40
infrastructuur	0.60	0.47	0.53	72
bereikb./beschikb. parkeerplek	0.51	0.45	0.48	49
parkeeroverlast	0.52	0.72	0.61	79
verkeersveiligheid	0.16	0.88	0.27	25
verkeersovertreding	0.08	0.12	0.10	8
schade en claims	0.00	0.00	0.00	64
burgemeester	0.00	0.00	0.00	3
hulp aan dak- en thuislozen	0.00	0.00	0.00	6
bijstand	0.00	0.00	0.00	1

Class	P	R	F1	Supp.
huur en verhuur	0.17	1.00	0.29	3
verhuizen	0.00	0.00	0.00	4
parkeervergunning/abonnement	1.00	0.57	0.73	7
parkeerkosten	0.67	1.00	0.80	2
documentaanvraag	0.00	0.00	0.00	1
hulpmiddelen en woonvoorz.	0.00	0.00	0.00	2
financiële regelingen	0.00	0.00	0.00	1
vaccinatie	0.00	0.00	0.00	1
luchtkwaliteit	0.33	1.00	0.50	1
parkeerterminal	0.00	0.00	0.00	1
parkeren bezoekers	0.67	1.00	0.80	2
overlijden	0.00	0.00	0.00	1
micro avg	0.44	0.52	0.48	2195
macro avg	0.32	0.39	0.30	2195

Table A.15: Llama 3.3 70B Instruct (two-step) – main topics, challenge test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.36	0.97	0.52	32
opruimen, afval en onderhoud	0.67	0.96	0.79	100
dagelijks leven en sociale geleg.	0.00	0.00	0.00	51
parkeren	0.73	0.65	0.69	17
vervoer	0.44	0.50	0.47	8
zorg	0.00	0.00	0.00	6
micro avg	0.56	0.66	0.61	214
macro avg	0.37	0.51	0.41	214

Table A.16: Llama 3.3 70B Instruct (two-step) – subtopics, challenge test set.

Class	P	R	F1	Supp.
criminaliteit	0.86	0.55	0.67	11
algemene veiligheid	0.17	0.82	0.29	11
overlast personen	0.22	0.67	0.33	6
geluidsoverlast	0.25	0.86	0.39	7
vuil/ongedierte overlast	0.10	0.83	0.17	6
onderhoud omgeving	0.43	0.78	0.55	54
buurt ambiance	0.00	0.00	0.00	3
evenementen, feestd. en herd.	0.00	0.00	0.00	1
hangjongeren	0.00	0.00	0.00	1
ophalen huiscont./grofvuil	0.81	0.33	0.46	40
verkeersmaatregelen	0.50	0.50	0.50	2
infrastructuur	0.29	0.50	0.36	4
parkeeroverlast	0.67	0.62	0.65	16

Class	P	R	F1	Supp.
verkeersveiligheid	0.08	1.00	0.15	2
verkeersovertreding	0.33	0.50	0.40	2
schade en claims	0.00	0.00	0.00	45
hulp aan dak- en thuislozen	0.00	0.00	0.00	1
parkeren bezoekers	1.00	1.00	1.00	1
micro avg	0.32	0.48	0.38	213
macro avg	0.32	0.50	0.33	213

A.1.5 BERTje (rule-based)

Table A.17: BERTje (rule-based) – main topics, full test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.87	0.90	0.88	515
opruimen, afval en onderhoud	0.92	0.93	0.93	799
dagelijks leven en sociale geleg.	0.61	0.62	0.62	136
parkeren	0.66	0.95	0.78	123
vervoer	0.66	0.95	0.78	99
zorg	0.88	0.58	0.70	12
financiële ondersteuning	0.25	0.50	0.33	2
wonen en ondernemen	0.18	1.00	0.31	2
burgerzaken	0.20	1.00	0.33	6
maatschappelijke ondersteuning	0.18	1.00	0.31	2
micro avg	0.82	0.90	0.86	1696
macro avg	0.54	0.84	0.60	1696

Table A.18: BERTje (rule-based) – subtopics, full test set.

Class	P	R	F1	Supp.
criminaliteit	0.87	0.81	0.84	280
algemene veiligheid	0.76	0.91	0.83	112
verlichting	0.79	0.98	0.87	82
overlast personen	0.73	0.92	0.81	75
geluidsoverlast	0.62	0.81	0.70	36
geuroverlast	0.51	0.86	0.64	49
vuil/ongedierte overlast	0.80	0.93	0.86	355
onderhoud omgeving	0.83	0.82	0.83	431
bruikb./beschikb. afvalcontainers	0.62	0.94	0.75	163
huisdierenoverlast	0.17	1.00	0.29	6
buurt ambiance	0.59	0.83	0.69	58
evenementen, feestd. en herd.	0.04	1.00	0.07	1
hangjongeren	0.62	1.00	0.77	10
ophalen huiscont./grofvuil	0.83	0.77	0.80	153
parkeergarage	0.81	1.00	0.90	15

Class	P	R	F1	Supp.
verkeersmaatregelen	0.61	0.85	0.71	40
infrastructuur	0.65	0.93	0.77	72
bereikb./beschikb. parkeerplek	0.57	0.86	0.68	49
parkeeroverlast	0.59	0.85	0.70	79
verkeersveiligheid	0.48	0.60	0.54	25
verkeersovertreding	0.26	0.75	0.39	8
schade en claims	0.72	0.33	0.45	64
burgemeester	0.29	0.67	0.40	3
hulp aan dak- en thuislozen	0.86	1.00	0.92	6
bijstand	0.00	0.00	0.00	1
huur en verhuur	0.18	1.00	0.31	3
verhuizen	0.25	0.75	0.38	4
parkeervergunning/abonnement	0.50	1.00	0.67	7
parkeerkosten	1.00	0.50	0.67	2
documentaanvraag	0.07	1.00	0.13	1
hulpmiddelen en woonvoorz.	0.18	1.00	0.31	2
financiële regelingen	0.00	0.00	0.00	1
vaccinatie	1.00	1.00	1.00	1
luchtkwaliteit	0.00	0.00	0.00	1
parkeerterminal	0.00	0.00	0.00	1
parkeren bezoekers	0.67	1.00	0.80	2
overlijden	0.50	1.00	0.67	1
micro avg	0.70	0.85	0.77	2195
macro avg	0.51	0.77	0.57	2195

Table A.19: BERTje (rule-based) – main topics, challenge test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.33	0.81	0.47	32
opruimen, afval en onderhoud	0.74	0.84	0.79	100
dagelijks leven en sociale geleg.	0.26	0.12	0.16	51
parkeren	0.56	0.82	0.67	17
vervoer	0.28	0.88	0.42	8
zorg	0.50	0.17	0.25	6
micro avg	0.52	0.64	0.57	214
macro avg	0.45	0.61	0.46	214

Table A.20: BERTje (rule-based) – subtopics, challenge test set.

Class	P	R	F1	Supp.
criminaliteit	0.26	0.73	0.38	11
algemene veiligheid	0.32	0.64	0.42	11
overlast personen	0.30	0.50	0.38	6

Class	P	R	F1	Supp.
geluidsoverlast	0.00	0.00	0.00	7
vuil/ongedierte overlast	0.06	0.50	0.10	6
onderhoud omgeving	0.50	0.52	0.51	54
buurt ambiance	0.00	0.00	0.00	3
evenementen, feestd. en herd.	0.33	1.00	0.50	1
hangjongeren	0.50	1.00	0.67	1
ophalen huiscont./grofvuil	0.53	0.23	0.32	40
verkeersmaatregelen	0.14	0.50	0.22	2
infrastructuur	0.22	1.00	0.36	4
parkeeroverlast	0.56	0.56	0.56	16
verkeersveiligheid	0.14	0.50	0.22	2
verkeersovertreding	0.00	0.00	0.00	2
schade en claims	0.80	0.09	0.16	45
hulp aan dak- en thuislozen	0.50	1.00	0.67	1
parkeren bezoekers	1.00	1.00	1.00	1
micro avg	0.30	0.38	0.34	213
macro avg	0.34	0.54	0.36	213

A.1.6 BERTje (LLM-predicted)

Table A.21: BERTje (LLM-predicted) – main topics, full test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.77	0.71	0.74	515
opruimen, afval en onderhoud	0.84	0.95	0.89	799
dagelijks leven en sociale geleg.	0.38	0.29	0.33	136
parkeren	0.76	0.79	0.77	123
vervoer	0.69	0.54	0.60	99
zorg	0.80	0.33	0.47	12
financiële ondersteuning	0.00	0.00	0.00	2
wonen en ondernemen	0.29	1.00	0.44	2
burgerzaken	0.71	0.83	0.77	6
maatschappelijke ondersteuning	1.00	0.50	0.67	2
micro avg	0.78	0.78	0.78	1696
macro avg	0.62	0.60	0.57	1696

Table A.22: BERTje (LLM-predicted) – subtopics, full test set.

Class	P	R	F1	Supp.
criminaliteit	0.78	0.03	0.05	280
algemene veiligheid	0.33	0.47	0.39	112
verlichting	0.83	0.88	0.85	82
overlast personen	0.22	0.17	0.20	75

Class	P	R	F1	Supp.
geluidsoverlast	0.27	0.94	0.42	36
geuroverlast	0.25	0.02	0.04	49
vuil/ongedierte overlast	0.58	0.59	0.58	355
onderhoud omgeving	0.68	0.49	0.57	431
bruikb./beschikb. afvalcontainers	0.46	0.44	0.45	163
huisdierenoverlast	0.24	0.83	0.37	6
buurt ambiance	0.24	0.14	0.18	58
evenementen, feestd. en herd.	0.04	1.00	0.07	1
hangjongeren	0.75	0.90	0.82	10
ophalen huiscont./grofvuil	0.44	0.58	0.50	153
parkeergarage	0.80	0.31	0.44	15
verkeersmaatregelen	0.33	0.07	0.12	40
infrastructuur	0.58	0.54	0.56	72
bereikb./beschikb. parkeerplek	0.28	0.24	0.26	49
parkeeroverlast	0.64	0.73	0.69	79
verkeersveiligheid	0.21	0.64	0.31	25
verkeersovertreding	0.10	0.12	0.11	8
schade en claims	0.45	0.31	0.37	64
burgemeester	0.67	0.67	0.67	3
hulp aan dak- en thuislozen	0.83	0.83	0.83	6
bijstand	0.00	0.00	0.00	1
huur en verhuur	0.29	1.00	0.44	3
verhuizen	0.75	0.75	0.75	4
parkeervergunning/abonnement	0.00	0.00	0.00	7
parkeerkosten	0.00	0.00	0.00	2
documentaanvraag	0.00	0.00	0.00	1
hulpmiddelen en woonvoorz.	1.00	0.50	0.67	2
financiële regelingen	0.00	0.00	0.00	1
vaccinatie	0.00	0.00	0.00	1
luchtkwaliteit	0.00	0.00	0.00	1
parkeerterminal	0.00	0.00	0.00	1
parkeren bezoekers	0.00	0.00	0.00	2
overlijden	0.33	1.00	0.50	1
micro avg	0.49	0.43	0.46	2195
macro avg	0.36	0.41	0.33	2195

Table A.23: BERTje (LLM-predicted) – main topics, challenge test set.

Class	P	R	F1	Supp.
veiligheid en omgeving	0.39	0.88	0.54	32
opruimen, afval en onderhoud	0.72	0.94	0.82	100
dagelijks leven en sociale geleg.	0.62	0.16	0.25	51
parkeren	0.65	0.65	0.65	17
vervoer	0.44	0.50	0.47	8
zorg	0.00	0.00	0.00	6
micro avg	0.60	0.68	0.64	214
macro avg	0.47	0.52	0.45	214

Table A.24: BERTje (LLM-predicted) – subtopics, challenge test set.

Class	P	R	F1	Supp.
criminaliteit	0.00	0.00	0.00	11
algemene veiligheid	0.31	0.82	0.45	11
overlast personen	0.30	0.50	0.38	6
geluidsoverlast	0.56	0.71	0.62	7
vuil/ongedierte overlast	0.10	0.83	0.18	6
onderhoud omgeving	0.55	0.43	0.48	54
buurt ambiance	0.67	0.67	0.67	3
evenementen, feestd. en herd.	0.50	1.00	0.67	1
hangjongeren	1.00	1.00	1.00	1
ophalen huiscont./grofvuil	0.67	0.40	0.50	40
verkeersmaatregelen	0.00	0.00	0.00	2
infrastructuur	0.14	0.25	0.18	4
parkeeroverlast	0.77	0.62	0.69	16
verkeersveiligheid	0.07	0.50	0.12	2
verkeersovertreding	0.00	0.00	0.00	2
schade en claims	0.67	0.09	0.16	45
hulp aan dak- en thuislozen	1.00	1.00	1.00	1
parkeren bezoekers	0.00	0.00	0.00	1
micro avg	0.38	0.38	0.38	213
macro avg	0.41	0.49	0.39	213

Appendix B

Appendix B

B.1 Confusion Matrices for BERTjeLLM (Full-Human Validated Test Set

B.2 Confusion Matrices for BERTjeLLM (Re-Annotated Test set)

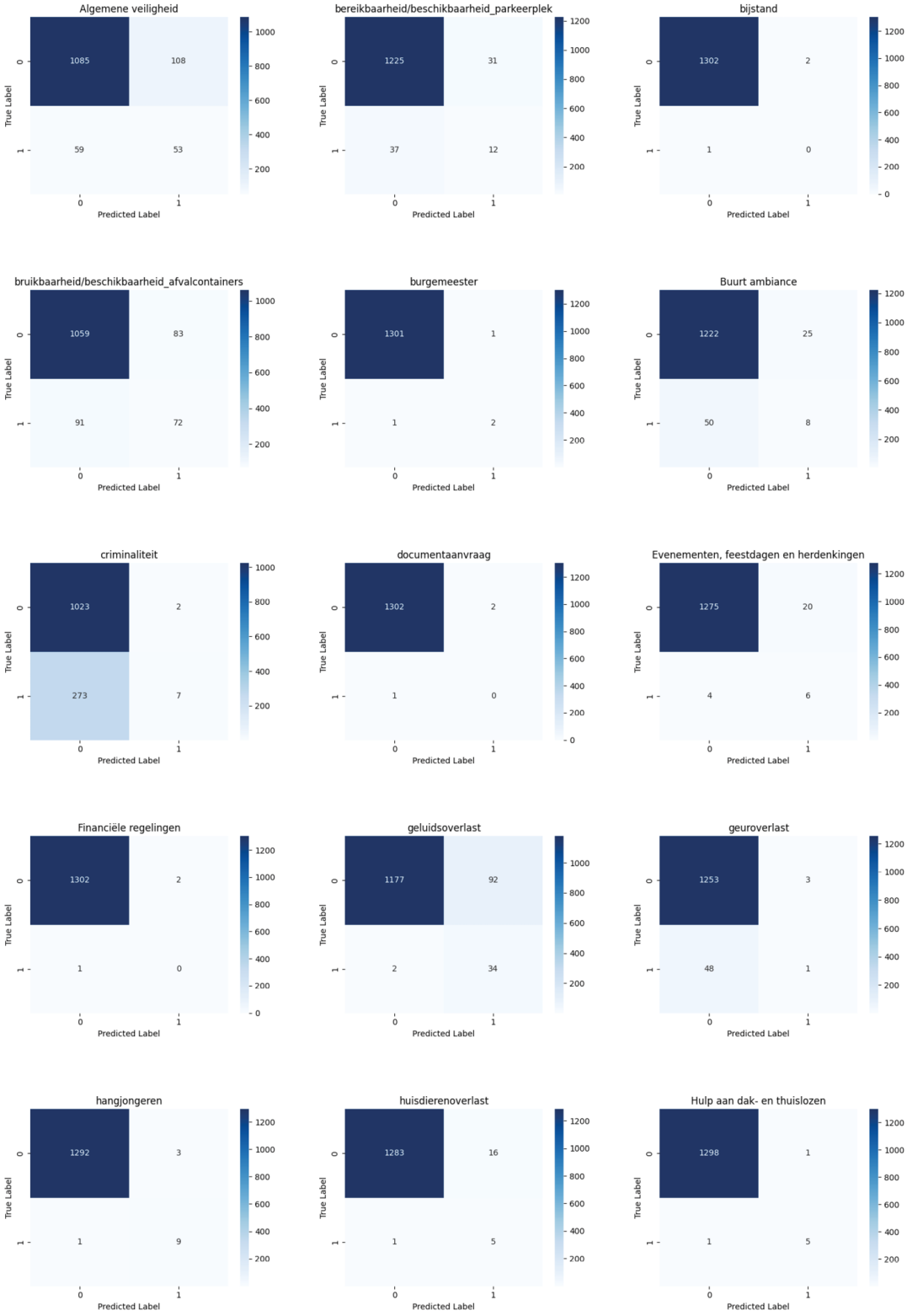


Figure B.1: Confusion matrices of subtopics for BERTjeLLM(I) (full test set)

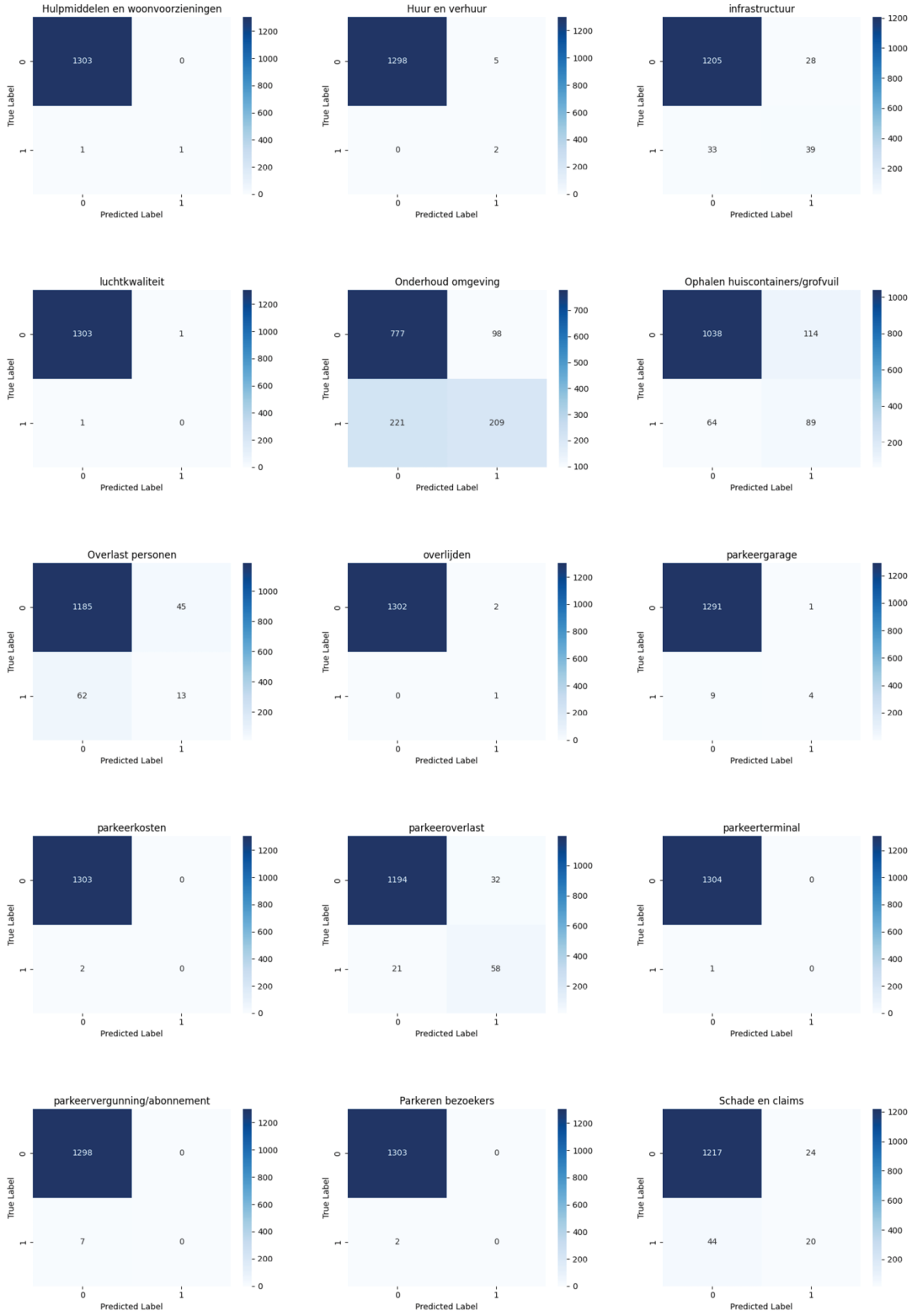


Figure B.2: Confusion matrices of subtopics for BERTjeLLM (II) (full test set)

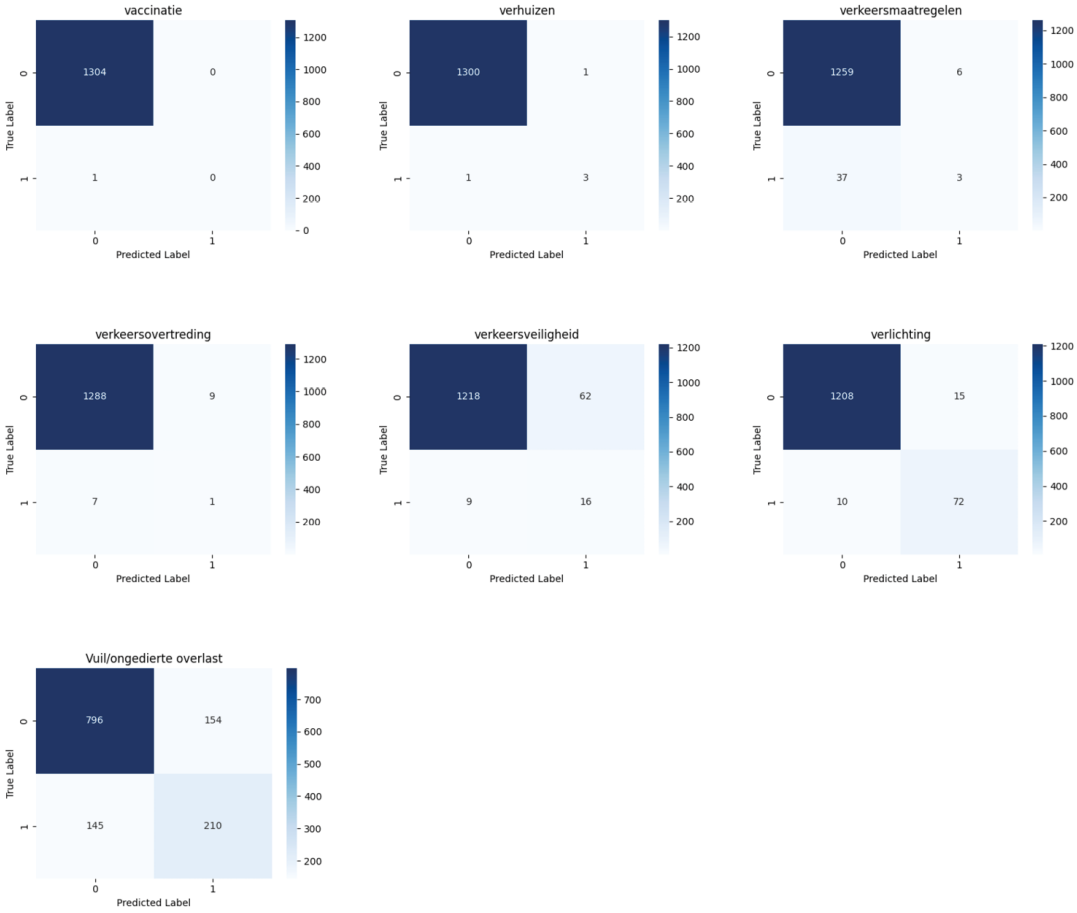


Figure B.3: Confusion matrices of subtopics for BERTjeLLM(III) (full test set)

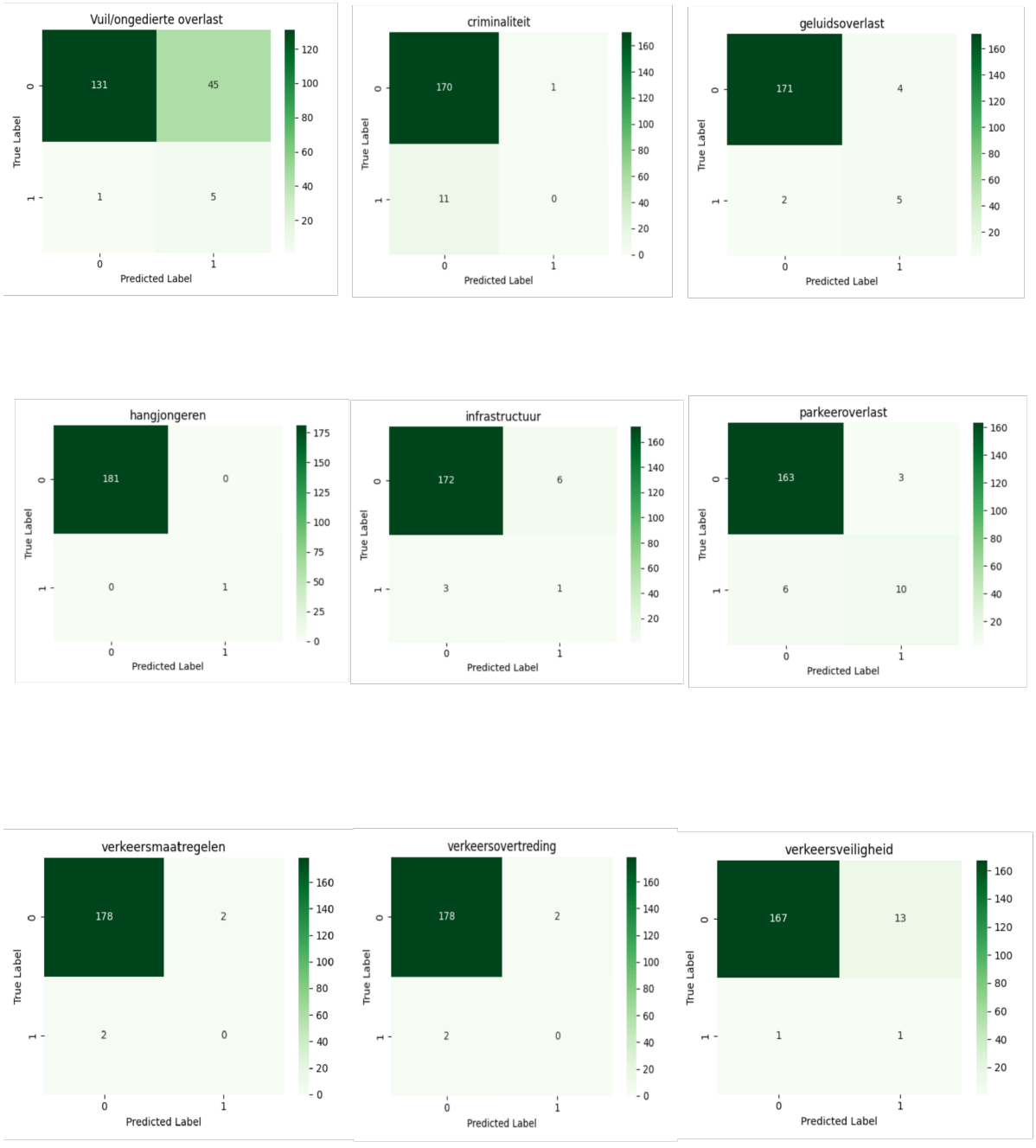


Figure B.4: Confusion matrices of subtopics for BERTje fine-tuned on LLM-predicted labels (I) (re-annotated test set)

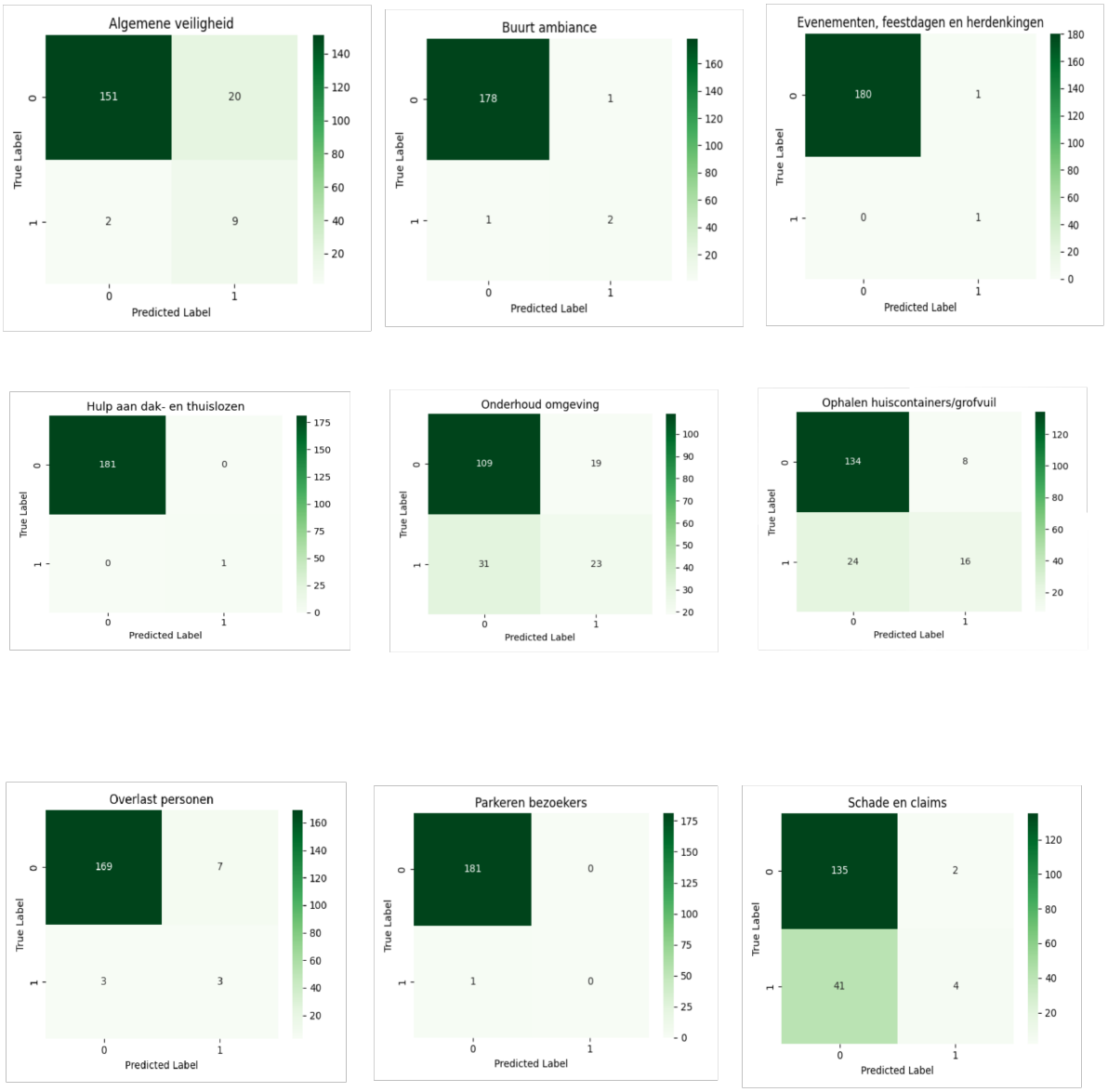


Figure B.5: Confusion matrices of subtopics for BERTje fine-tuned on LLM-predicted labels (II) (re-annotated test set)

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